LODES Design and Methodology Report: Methodology Version 7

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Abstract

The purpose of this report is to document the important features of Version 7 of the LEHD Origin-Destination Employment Statistics (LODES) processing system. This includes data sources, data processing methodology, confidentiality protection methodology, some quality measures, and a high-level description of the published data. The intended audience for this document includes LODES data users, Local Employment Dynamics (LED) Partnership members, U.S. Census Bureau management, program quality auditors, and current and future research and development staff members.

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Disclaimer: Any opinions and conclusions expressed herein are those of the authors and do not represent those of the U.S. Census Bureau, the FDIC, or the United States. Much of the work for this document was done while Matthew Graham and Mark Kutzbach were employees of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection. (Project 7515812: CBDRB-FY25-0223)
Acknowledgments: Numerous individuals and groups have contributed to the development and dissemination of LODES. Primary among them was Matthew Graham, who passed away before we ever completed this document. LODES was his pride and joy, both in its simplicity and usefulness, as well as the underlying complexity. He wrote much of this document, and developed most of the code for the data product in its current form. He was an incredible public servant and thoughtful coworker.
We also want to acknowledge the work that innumerable people did to help develop LODES, and the team that continues to produce LODES now.

The core difficulty is that everything is very complex.

ANONYMOUS

Selected by Matt Graham

Contents

1 Introduction

1.1 Purpose and Audience

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1.2 Currency

This document covers data development and release through Version 7 of LODES, which includes data from 2002 through 2018. Note that LODES code versions are not numerically synchronized with LODES Dataset Structure Formats.[1] The most recent "release" revision of the LODES processing codebase is 07.01.012.

1.3 Program Overview

The Longitudinal Employer-Houshold Dynamics (LEHD) Program is an innovative statistical program within the U.S. Census Bureau's Center for Economic Studies. The Program uses modern statistical and computing techniques to combine federal and state administrative data on employers and employees with core Census Bureau censuses and surveys in order to produce useful statistics on the US economy and job market all while protecting the confidentiality of people and employers who provide the data. The LEHD Program accesses state administrative data via the Local Employment Dynamics (LED) Partnership, which is a voluntary federal-state partnership that was started in 1999. Some of the public-use data products released by the LEHD Program include Quarterly Workforce Indicators (QWI)[2], Job-to-Job Flows (J2J)[3], and LODES[4]. The Census Bureau launched LODES in February 2006 with the Cornell VirtualRDC hosting the data for public download.

[1] See U.S. Census Bureau [2020b]

[2] For more on the processing of jobs data and production of QWI, see `http://lehd.ces.census.gov/data/#qwi` and Abowd et al. [2009].

[3] For more on the processing of jobs data and production of J2J, see `http://lehd.ces.census.gov/data/j2j_beta.html`.

[4] For more on the released data files of LODES, see `http://lehd.ces.census.gov/data/#lodes`.

LODES is widely used by federal, state, and local governments as well as by businesses, nonprofits, and academics for economic development, transportation planning, and research and has been credited as an innovation in statistics and dissemination. Comparability to the American Community Survey (ACS) Commuting data was assessed in Graham et al. [2014] and Green et al. [2017], and measures of total variance for workplace tabulations were made in McKinney et al. [2017] and McKinney et al. [2020] (for more on this last paper, see also Section 5.2 and Appendix A).

These data have been used to understand the following areas of research and policy: the spatial distribution of urban employment and commuting (see Manduca [2018], Shearer et al. [2019], and Schleith et al. [2019]), the dynamics of transportation, land-use, and access (see Horner and Schleith [2012], Johnson et al. [2015], Davidson and Ryerson [2018], and Karner [2018]), disparities in access to jobs and the effect of spatial mismatch on labor market outcomes (see Andersson et al. [2018], Kneebone and Holmes [2015], and Chen et al. [2019]), the potential impacts of future disasters from various perspectives (see Kermanshah and Derrible [2016], Jiang et al. [2020], Boeing [2018], and Boeing [2018]), the effects of changing employment policies (see Shoag and Veuger [2016] and Clifford and Shoag [2016] and Ballance et al. [2020]), and the relationship between housing and commuting dynamics (see `https://www.zillow.com/research/reverse-commuters-urban-suburban-27168/`, `https://ggwash.org/view/77446/three-graphs-and-two-maps-help-us-understand-the-coronavirus`, `https://www.zillow.com/research/facebook-ipo-home-values-22940/`, `https://www.redfin.com/news/introducing-opportunity-score/#how`, and `https://www.dllr.state.md.us/lmi/wiacommuting/`).

2 Project Design

2.1 Overview

The LEHD Origin-Destination Employment Statistics (LODES)/OnTheMap project encompasses two major efforts. The first is the research, development, and production of the LODES data product. This is an origin-destination dataset that connects employment and residential locations for workers and tabulates them to the census block level of geography. This dataset includes worker and employer characteristic information such as worker age, job earnings, and employer industry. More detail on the exact specifications of the output datasets can be found in the Dissemination section.

The second piece of the project is the development and release of an online data analysis tool called "OnTheMap."[5] Due to the size and spatial complexity of the LODES data, this online tool - a visual, mapping-based application - was developed in order to aid users in their data queries and analyses and to grow the user base of the data beyond expert data analysts. OnTheMap serves as one of the primary dissemination methods for the LODES datasets.

This Quality Profile will primarily cover the LODES data processing system; detailed documentation of the OnTheMap application is available at `http://lehd.ces.census.gov/applications/help/onthemap.html`.

2.2 Similar Data Products

Users may generate statistics comparable to LODES from several other public use data products. Any comparisons of levels or trends should consider differences in coverage, definitions, data processing, and confidentiality protection methods. For a detailed comparison of LODES and the American Community Survey (ACS), see Graham et al. [2014]. For a comparison of commuting statistics for job records linked across the LODES and ACS input files, see Green et al. [2017].

Both the Census Transportation Planning Package (CTPP) and the ACS Commuting statistics provide counts of home-to-work commuting flows. The CTPP was created by the Census Bureau for the transportation community as a special tabulation of the 1990 and 2000 Decennial Census Long Forms and more recently as a special tabulation of the ACS. At its finest geographical resolution, the CTPP published an origin-destination matrix at the census tract level.[6]

Commuting statistics (or Journey-to-Work) are tabulated from the ACS, an ongoing survey of the Population of the United States conducted by the U.S. Census Bureau. The ACS questions related to travel focus solely on commuting and do not ask about nonwork travel. Respondents answer questions about where they work, what time they leave home for work, the means of transportation used to get there, the number of workers riding in the

[5] OnTheMap is a registered trademark of the U.S. Census Bureau and can be accessed at `http://onthemap.ces.census.gov/`.

[6] See `http://www.fhwa.dot.gov/planning/census_issues/ctpp/` and `http://ctpp.transportation.org/Pages/default.aspx`

car, truck, or van, and how long it takes to travel to work, as well as whether they telework or work from home.. Along with several other questions on housing and person characteristics, these travel-related questions previously appeared on the decennial census long form, but were moved to the ACS after the 2000 Census. The decennial census now produces only a count of the nation's population and a snapshot of its most basic demographic and housing characteristics.[7][8]

The LODES workplace margin may be compared with tabulations of employer data including QWI, the Quarterly Census of Employment and Wages (QCEW) and the County Business Patterns and ZIP Code Business Patterns. The LODES residence margin may be compared with ACS tabulations of the employed, civilian population. Caveats may apply to these comparisons, as stated in previously cited research.

While Census Bureau data products and the CTPP focus on aggregate commuting, the Federal Highway Administration's (FHWA) National Household Travel Survey (NHTS) describes travel behavior (including both work and non-work trips). Conducted since 1969, with the most recent survey in 2017, the NHTS "provides data on individual and household travel behavior trends linked to economic, demographic, and geographic factors that influence travel decisions and are used to forecast travel demand." The 2017 NHTS had a target size of 129,112 households, drawn as a random sample with additional representation from sponsoring states and agencies.[9]

3 Data Frame and Sources

LODES is based upon a job frame, unlike other data products from the Census Bureau, which may be based upon household, business, or person frames.[10] This section describes the frame of jobs that the LODES data represent and describes the various data sources that contribute to LODES. Please note that this section borrows extensively from Graham et al. [2014].

3.1 Frame Definition

3.1.1 Collection

LODES data are generated from a variety of administrative and survey data sources. There is no new respondent burden required to produce this data product. The core jobs data come from states that are part of the LED Partnership. States joined at different times and the extent of time series availability varies by state. By 2013, all states, the District of Columbia, Puerto Rico, and the U.S. Virgin Islands had joined LED. These records are supplemented by earnings records of federal workers from the Office of Personnel Management (OPM). A detailed description of data sources is given in Section 3.2.

3.1.2 Coverage

LODES provides counts of unemployment insurance covered wage and salary jobs, as reported by state labor market information offices and by OPM (for 2010 onwards). The state data, covering employers in the private sector and state and local government, account for approximately 97 percent of civilian wage and salary jobs, as estimated by the Bureau of Labor Statistics.[11] Partner states submit two quarterly files from their administrative data systems. Wage record files list the quarterly earnings for workers at each job they hold in a pay period falling within the quarter. While files from a few states have information on hours worked, these data are being researched and neither QWI nor LODES currently use them. Quarterly Census of Employment and Wages (QCEW) files, formerly known as the ES-202 program, list the establishments at employers active in a quarter, and report location, industry, ownership type, and workforce size. LEHD does not receive or process Multiple Worksite Reports (listing the establishment associated with an employer), which states have already incorporated into QCEW files. The OPM data cover about half of federal government civilian employment.[12] Within this universe of jobs, the LODES data have almost full coverage,

[7]This paragraph has been excerpted from Graham et al. [2014], which compares the designs and methodologies of LODES and ACS Commuting Data.

[8]See http://www.census.gov/topics/employment/commuting.html for more information about the ACS Commuting data.

[9]See Westat [2019] for more information.

[10]For more information on the status of the Job Frame, refer to Foote et al. [2023].

[11]See early discussion in Stevens [2007] and more recent coverage statistics in Robertson [2017] and U.S. Bureau of Labor Statistics [2019].

[12]Federal employment first appears in 2010 LODES data. For more information on coverage within the civilian workforce, see Data Definitions: http://www.fedscope.opm.gov/datadefn/aboutehri_sdm.asp#cpdf3 and http://fedscope.opm.gov/datadefn/index.asp. Additional information on coverage of the OPM data will be provided in a forthcoming technical

although some in-universe firms may not report during a particular quarter.[13] Examples of job types beyond the scope of LEHD earnings records are: the military and other security-related federal agencies, postal workers, some employees at nonprofits and religious institutions, informal workers, and the self-employed.

3.1.3 Geographic and Longitudinal Scope

LODES are published as an annual cross-section from 2002 onwards, with each job having a workplace and residence dimension. States provide data for employers located within the state, so only states contributing data to LEHD in a given year will be included in the workplace dimension for that year. As of the writing of this document, only Alaska, Michigan and Missouri are out of partnership, and the data also include Puerto Rico. The federal administrative data providing residence locations are national and do not depend on a state joining the LED Partnership. LODES does not currently release residence data for Puerto Rico and the U.S. Virgin Islands. While the domain of workplace locations varies across years, all states (except Puerto Rico and the U.S. Virgin Islands) have some - possibly incomplete - residence data from 2002 onwards.

3.1.4 Job Definition and Reference Period

The LEHD infrastructure collects earnings records and QCEW data quarterly. For a job to be included in LODES, a person must have positive earnings from a firm in both the reference quarter (Quarter 2 of each year: April to June) and the previous quarter (Quarter 1: January to March). Jobs with earnings in adjacent quarters are inferred to have been held at the "seam" of the quarters, in this case, April 1. Because quarterly earnings reports often depend upon completion of a pay period during a quarter, LODES may underrepresent very short jobs held for just one pay period[14] The LODES job count corresponds to the QWI indicator "beginning-of-quarter" jobs, for the second quarter.

While all jobs in the LODES data frame are presumed to be held on April 1, some job characteristics may have a different timeframe. Earnings for a job include all income from an employer during all pay periods falling within the quarter. LODES uses second quarter nominal earnings to classify jobs into low, middle, and high earnings categories. LODES further classifies jobs as "primary/dominant" if the job earned the person holding the job the most earnings of all beginning-of-quarter jobs held by the person during the reference period. Under the "primary/dominant" definition, there is a one job per worker equivalency. Other employer characteristics, including ownership, industry, and firm size and age, are also based on the employer report in the second quarter. The timing of the residential location is not as precise, but use of tax filing data means that many residential locations have a reference date on or before April 15 of each year. (Graham et al. [2016]) Worker demographics are assumed to be time invariant (excepting age, which changes in a consistent way between release years), and are based upon survey and administrative records. Educational attainment is not available for those in the youngest age category (aged 29 or younger).[15]

3.1.5 Job and Worker Characteristics

LODES tabulates jobs by workplace and residence location, as well as by a set of job, employer, and worker characteristics. All tabulations are available by job dominance (2 categories) and employer ownership (3 categories). See the "Job Definition" section above for further explanation of job dominance. Ownership includes federal government, state and local government, and the private sector, although the federal jobs are available only for 2010 and later.[16] Other characteristics are industry (3 industry segments, 20 NAICS industry sectors),[17] firm size (5 categories), firm age (5 categories), average monthly earnings (3 categories), age (3 categories), sex (2 categories), race (6 categories), ethnicity (2 categories), and educational attainment (4 categories, and not applicable for those aged 29 and under).[18]

document.

[13]If firms do not report for a single quarter, these data holes are addressed by the application of the EHF impute. For more details, see `https://lehd.ces.census.gov/data/lehd-snapshot-doc/S2023/sections/jobs_level/ehf.html`

[14]For more discussion on the number of single-quarter jobs, consult Hyatt and Spletzer [2015].

[15]QWI publishes statistics by categories of educational attainment for workers age 25 and older. Educational attainment is not tabulated for those under 25 because too many of those workers had incomplete information on educational attainment. The original development of LODES set 29 and under as one of the age categories, so LODES does not tabulate educational attainment for that entire category.

[16]In OnTheMap, ownership is made available in two categories: all jobs and private-sector only jobs. In the raw LODES datafiles, federal only jobs are published as an additional file.

[17]NAICS is the North American Industrial Classification System. More information can be found at `https://www.census.gov/naics/`.

[18]Additional detail on the variables provided in LODES can be found in U.S. Census Bureau [2020b]

Not all characteristics are available throughout the time series. Sex, race, ethnicity, and educational attainment are available beginning in 2009. Firm size and age are available beginning in 2011. Federal data from OPM begin in 2010. Earnings categories are for nominal values that are constant over the span of the series. Three age categories are not aligned with the eight age categories in QWI.

3.1.6 Location definitions (workplace and residence)

LODES documentation often uses the terms “origin-destination flow” and “commute” to refer to the residence and workplace that are tied to a job. However, these flows may not always represent a trip pattern. As is discussed below, both the workplace and residence may differ from a typical travel survey concept. The LEHD data infrastructure and LODES in particular have no information about transport mode (if any), and a worker may telework from home or another site regularly, or work only infrequently.

For LODES, a place of work is defined by the physical or mailing address reported by employers in the QCEW (formerly ES-202) or Multiple Worksite Reports.[19] An address from administrative data may or may not be the actual location that a worker reports to most often. The distinction of worksite and administrative address may be especially significant in some industries such as construction, where work is often carried out at temporary locations. In some cases, employers do not provide a multiple worksite report when it would be appropriate to do so. Nonreporting of multiple worksites is especially common with state and local governments and school districts. In such a case, LEHD infrastructure files assign all workers for that employer (within the state) to the main address provided. BLS data shows a national non-compliance rate of 5.6 percent of multi-unit employers responsible for about 4.45 percent of multi-unit employment.[20]

Residential information in LODES is sourced from the Federal administrative data from a number of different programs (Graham et al. [2016]). Because different programs have different collection periods and different uses for addresses, we may receive multiple different residential locations within a year for a single person. Individuals may have multiple valid residential addresses within a year, corresponding to permanent or seasonal moves. Current versions of processing attempts to choose the “best” location based on proximity to the reference period (Q2), reliability of the source, and agreement among sources both within the year and across years. In cases where residential information is not available or is not complete, imputations are performed as described below.

3.2 Data Sources

This section outlines the data sources that are used in the LODES processing system. Overall, these data sources fall into four different categories: LEHD Infrastructure Files, Place of Residence, Decennial Census Products, and Geographic Information. Tables 1-4 provide descriptions of the input datasets, the years used, their purpose in LODES processing, and citations or links for more information. Maintaining the currency and compatibility of these inputs is a significant challenge. The LEHD Program must reguarly evaluate the quality and definitions of inputs to ensure that existing processing methods apply. Delays in the availability of critical inputs, or changes to those inputs, have sometimes resulted in delays in the release of LODES. In the best circumstances, LEHD can release LODES no sooner than 18 months after the reference date of economic activity (April 1 of each year). As an extreme case, a disruption to the availability of place-of-residence data delayed the release of LODES data for 2012 and 2013 until 2015.

3.2.1 LEHD Infrastructure Files

LODES processing is “downstream” of quarterly processing of the LEHD Infrastructure Files. (Graham et al. [2022]) The LEHD Production team updates LODES inputs including the Person History File (PHF),[21] Employer Characteristics File (ECF), and Individual Characteristics File (ICF). The LEHD data vintaging system allows LODES to maintain records of the exact copies of those files, and the code used to produce them, that contribute to each release of LODES.

[19] The Bureau of Labor Statistics provides links to all states’ Multiple Worksite Reports at the following address: `http://www.bls.gov/respondents/mwr/forms/mwr-forms.htm`

[20] See Spear [2011, Table 2.1].

[21] The Person History File (PHF) and the Job History File (JHF) are two names for the same data in the LEHD Infrastructure.

Table 1: LEHD Infrastructure Files

Person-level Files

Name	Years Used	Description and Purpose	For More Information
Person History Files (PHF)	2002-Current	Job information (person-employer relationship), including earnings	See Abowd et al. [2006].
Individual Characteristics File (ICF)	2002-Current	Person Characteristics (including implicates files for age, sex, race, and ethnicity)	Ibid
ICF OPM	2010-Current	Person Characteristics for Federal employees covered by Office of Personnel Management (OPM)	See Vilhuber and McKinney [2014, Section 10].

Employer Files

Name	Years Used	Description and Purpose	For More Information
Employer Charactistics File (ECF)	2002-Current	Information at establishment and firm level, including ownership, industry, location, firm age and size.	Ibid.
QWI Establishment Weights File	2002 - Current	Job Weighting Factors at state-level, in order to match to more closely align beginning of quarter employment from UI wage records to total reference period employment from QCEW	Ibid.

3.2.2 Place of Residence

LODES uses annual files of place-of-residence data that were historically produced elsewhere in the Census Bureau, but have more recently been developed within the LEHD Program. The LEHD Program received the Composite Person Record (CPR) for the years 1999-2010. With the discontinuation of CPR, the LEHD Program worked with the CES data staff to produce an integrated version of the Master Address File-Auxiliary Reference File (MAF-ARF) for 2011. For 2012 onward, the LEHD Program has developed a residence file customized for the needs of producing employment data, known as the Residence Candidate File (RCF)[22]

3.2.3 Decennial Census Data

LODES uses public-use data from the Decennial Census as a prior assumption for the spatial distributions. The first year LODES was produced for each state, the distribution of commutes from the 2000 CTPP was used as a prior assumption for the synthetic data model. Because the spatial precision of the synthetic data model varies by distance according to a coarsening schedule (specified in the domain), residence distributions in 2000 and 2010 Census are used to further disaggregate jobs, by residence, to the Census block level. The switch from 2000 Census to 2010 Census first occured for production of 2011 LODES. For more information on coarsening, see Section 4.

3.2.4 Geography

Additional geographic data are necessary to support the various edits, imputations, and protection methods that are part of the LODES processing system. A full list of these inputs are given in Table 4. Some of the basic operations supported by these datasets include:

- LODES data are longitudinal but must be published in a single geographic frame, therefore some datasets are used to transform (or recode) historical geography to current boundaries.
- Some data processing functions rely on hierarchical relationships between statistical entities, such as census tracts and public-use microdata areas (PUMAs). In these cases datasets are constructed from appropriate reference files to establish the necessary relationships among geographic entities of varying scales. These kinds of relationship files are often referred to as “crosswalks.”

[22] See Graham et al. [2015, 2016, 2017] for more details.

Table 2: Place of Residence

Name	Years Used	Description and Purpose	For More Information
StARS Composite Person Record (CPR)	2002-2010 (superseded)	Compiles national-level administrative datasets for internal Census purposes; and is used to attach residential addresses to the person-job information from PHF.	See Farber and Leggieri [2002], Berning and Cook [2003], and Bye and Judson [2004].
MAF-ARF	2011 (superseded)	An amendment to MAF using administrative records to link PIKs to MAF IDs to attach residential addresses to person-job information.	For brief summaries, see `https://www2.census.gov/programs-surveys/research/guidance/nlms/maf-description.pdf` and U.S. Census Bureau [2020a] as well as the discussion of the MAF-ARF in Finlay [2016].
Residence Candidate File (RCF)	2012-Current	RCF is internal LEHD replacement for CPR and MAF-ARF as source of residential location information. Federal administrative sources with reseiential information are combined in this process to define the candidate set of geographies (and weights) for each individual.	See Graham et al. [2015, 2016, 2017].

Table 3: Decennial Census Data

Name	Years Used	Description and Purpose	For More Information
Census Transportation Planning Package (CTPP)	2000	Special Tabulation of 2000 Long Form that includes information about residence, workplaces, and origin-destination relationships. LODES specifically uses the OD relationships as the public prior to protect residence in the 2002 data year.	`http://ctpp.transportaion.org/Pages/default.aspx`
2000 Decennial Census, Summary File 1 (SF1)	2000 (superseded)	LODES uses P1 (Total Population) and P8 (Hispanic/Latino by Race) tables at the Census block to decoarsen the synthetic data.	`https://www.census.gov/prod/cen2000/doc/sf1.pdf`
2010 Decennial Census, Summary File 1 (SF1)	2010	LODES uses P1 and P5 Tables (same as above), for decoarsening of synthetic data.	`http://www.census.gov/prod/cen2010/doc/sf1.pdf`

Table 4: Geography

Name	Years Used	Description and Purpose	For More Information
Master Address File (MAF)	2002-Current	LODES uses MAF as a reference dataset for looking up residential and business addresses for block-level geocoding.	`https://www.census.gov/programs-surveys/geography/about/partnerships/decennial-partnerships/2010/luca.html`
TIGER/Line Data	2010 and current year[23]	LODES uses shape files and relationship files in order to provide linkages between different geographic levels, as well as provide internal points for geography between which distance and direction can be calculated (for commute distance)	`https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html`

Table 5: LODES Processes

Data Integration	Disclosure Avoidance	Quality Assurance	Public Release
EXT	SYNTHPREP	OTMQA	OTMPU
WHATA	SYNTHESIZE	QA steps throughout other processes	OTMAPP
USDOM			OTMREL
WHATB			

- Some data processing functions require a complete set of geographic entities at different levels, either to define a domain or to check for invalid values.

In some cases the geographic inputs need only be produced once or once every decade (immediately following the redefinition of census blocks as part of the Decennial Census), however some of the files - particularly the crosswalks that support the public-use file creation - must be updated each year as the U.S. Census Bureau updates its public geographic data product.

4 Methodology

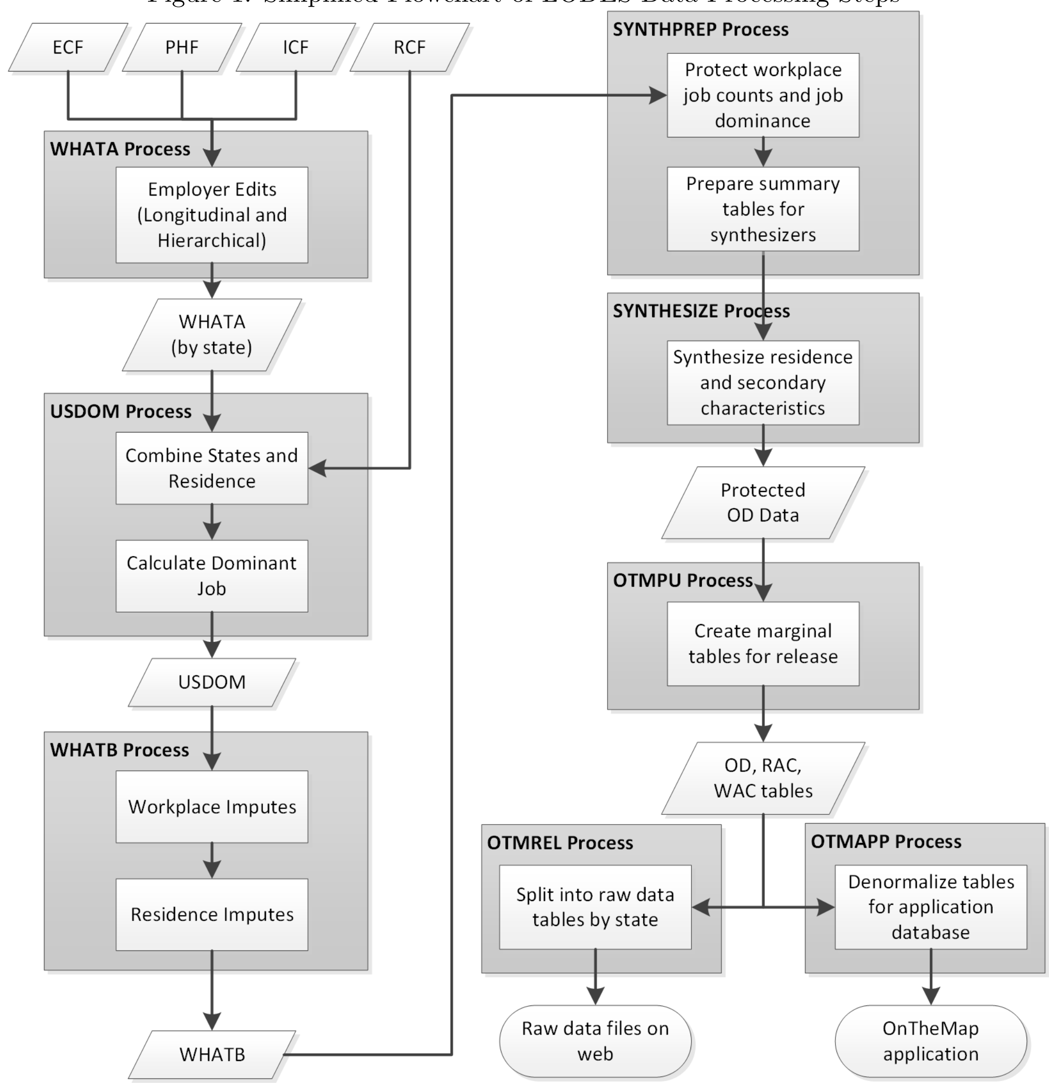
This section outlines the LODES production process and describes edits and imputations and confidentiality protection measures.

4.1 Data Production Process Overview

There are four main components of the LODES production process: Data Integration, Disclosure Avoidance, Quality Assurance, and LODES Release. Data Integration brings together inputs and applies edits and imputations resulting in a confidential jobs file with completed information on all characteristics. Disclosure Avoidance transforms the jobs data into values that represent the patterns of the underlying data, but limit inferences about sensitive characteristics. Quality Assurance occurs both as a dedicated process and during each of the other processes. The Release stage prepares the protected data for public dissemination as tables as well as through the OnTheMap web tool. The descriptions in the following sections sometimes refer to the names of processes as implemented in the codebase. Each process includes a series of modules. In the processing system, the data output of a process is often known by the same name as the process. The names of the processes are listed in Table 5 within their respective processing components and Figure 1 visually summarizes the process workflow and major functions.

[23] See `http://lehd.ces.census.gov/applications/help/onthemap.html#!geographic_data` for the latest vintage of geographic data.

Figure 1: Simplified Flowchart of LODES Data Processing Steps



4.1.1 Data Integration

Data Integration includes a set of auxiliary programs captured in the EXT process. The modules in this process perform preparatory tasks including updating files to more recent vintages of tabulation geography (e.g. 2000 to 2010), summarizing Decennial Census data, and calculating distances.[24] Some of these tasks need not be re-run annually, in which case, output files are used for multiple years of processing. The EXT process is responsible for producing the coarsened domain (see below for more detail), or the set of all residence locations eligible for flows to a workplace Census tract.

The WHATA process creates a snapshot of jobs for each source of earnings records. Along with identifying

[24]The end points used for calculating distances are "internal points" for each geographic entity. Internal points simply mean a point inside the boundary on an entity (census block, tract, etc.), which is sometimes not the same as a mathematical centroid, when the mathmatical centroid would fall outside the shape boundaries. The distance calculation itself employs Vincenty's inverse method (Vincenty [1975]), which is an iterative method for calculating geodesic distances on an ellipsoid. The convergence problem with this method does not apply to LODES because no pair of internal points within the United States and its terrories are close to being antipodal, and in practice the method usually converges within a few iterations. In addition this method also provides azimuths (direction from one endpoint to face the other endpoint), which are used for quality evaluations (see Section 5.2) and by the OnTheMap application (see Section 6.5).

jobs held on a reference date from the PHF, the process gathers all the necessary job information (PHF), including characteristics of workers (ICF), employers (ECF), and employer weights (QWI). The PHF, ICF, ECF, and QWI processes are all standard LEHD data infrastructure processes as discussed in Section 3.2. The only modifications to the infrastructure files at this point in the process are to aggregate cases of multiple earnings records referring to the same job and to complete missing employer characteristics through either hierarchical or longitudinal edits.[25]

The USDOM process then stacks together all the state snapshots of jobs and defines the dominant job for an individual as the one with the highest earnings at the beginning of the reference quarter (quarter 2). The process joins in place-of-residence data from the RCF at the person level and applies flags based on the accuracy and precision of geographic information on workplace and residence. Lastly, the process applies formatting codes to job characteristics that will be used for edits and imputations as well as eventual tabulation.

The WHATB process ensures item completeness of workplace and residence information, two characteristics that are essential for the basic margins by which LODES releases data. The process also applies definitions from the EXT process for geographic aggregation, or coarsening. The output from the WHATB process is the best, most complete version of the confidential data, and any published data can be measured against it for quality analysis. Edits and imputations are described in more detail in Section 4.2.

4.1.2 Disclosure Avoidance

The SYNTHPREP process defines the protected job count by workplace as well as by job dominance and prepares the distributions needed for protecting secondary characteristics and synthesizing place of residence data. SYNTHPREP draws on the WHATB, LODES data from the previous year, and outputs of the EXT process to produce the necessary counts and distributions. This process makes use of confidential parameters in order to protect workplace counts.

The SYNTHESIZE process draws secondary characteristics and residences from distributions produced in SYNTHPREP according to counts of jobs also produced in SYNTHPREP. This process creates a job-level origin-destination matrix, as well as firm age and size at each workplace. Lastly, it decoarsens synthesized residence locations to a fine level of geographic granularity, the Census block. Disclosure avoidance methods are described in more detail in Section 4.3.

4.1.3 Quality Assurance

The OTMQA process operates throughout data production and includes modules to validate production outputs. After the synthetic data are created, they are checked both against WHATB and longitudinally in order to ensure proper data quality; any issues found are recorded and, if possible, corrected before release. The details of Quality Assurance procedures are discussed in Section 5.

4.1.4 Public Release

Using the synthetic outputs from the SYNTHESIZE process, OTMPU creates three marginal tables that form the basis of the public-use datasets: Origin-Destination, Residence Area Characteristics, and Workplace Area Characteristics files. It also creates a "job type" variable, which conflates job dominance and ownership.[26] OTMREL breaks this into state-based chunks for release as raw data files, while OTMAPP creates all the inputs for the OnTheMap web application, interacting the public-use files with a full hierarchical geography.

4.2 Edits and Imputations

To help users understand the underlying quality of the data in the LODES system, this section decribes the edits and imputations applied during the LODES processing system, as well as some weighting procedures.[27] Most of the edits and imputations are used to fill in missing data (item missingness rather than unit missingness) and to provide some data with more geographic detail. Before the synthetic protection methods (described in Section 4.3) can be applied, all variables must be complete, including geography at the 2010 census tabulation block level. Each edit and imputation process is run through a series of tests when it is first developed by LEHD researchers. Ongoing tests

[25] Hierarchical edits occur when an establishment (SEINUNIT) is missing information on an establishment characteristic; in this case, the missing information is taken from the employer (SEIN). Longitudinal edits occur for an establishment when the information that it is missing can be imputed from adjacent quarters.

[26] Detailed documentation of the public-use LODES files can be found in U.S. Census Bureau [2020b].

[27] For the purpose of this document we differentiate edits and imputations as follows: edits are deterministic, whereas imputations include a stochastic process.

Table 6: Edits and Imputations

Process	Edit/Imputation
WHATA	Successor-Predecessor Edits
	Establishment Characteristics Edits
	Record Edits
WHATB	Workplace Imputations
	Residential Imputations

of the edits and imputations are coded in QA modules that accompany each process. More information regarding the quality measures of the edits and imputations can be found in Section 5. Table 6 lists all the major edit and imputation procedures as well as the process in which they are implemented.

4.2.1 Edit Procedures

Successor-Predecessor Edits

The first edit that occurs in the WHATA process is the "successor-predecessor" edit. Similar versions of this edit are also implemented in the LEHD infrastructure processes.[28] The edit identifies jobs affected by an identifier transition, such as when an employer splits into multiple new business entities and some or all workers continue in their employment with these new entities, or when an employer begins reporting all their employees under a new SEIN.

The successor-predecessor edit links the successor firm(s) to the predecessor firm(s) so that an individual worker's single job spell, which happens to break across the organizational transition, can be recongnized as such. If the successor-predecessor edit were not applied, a single job may appear as two (or more) jobs with two (or more) firms.

An employee must be employed by the same employer in the first two quarters of the year to qualify as having a "Beginning-of-Quarter" job in Quarter 2 and hence be counted in LODES. In the case of an organizational transition as described above, jobs in the reference quarter that have the same predecessor employer are combined for purposes of calculating earnings and establishing job continuity; establishment assignments are derived from the linked jobs with the highest earnings in the reference quarter.

This edit increases the count of beginning-of-quarter jobs in the case of successor-predecessors that are non-overlapping in the first and second quarter. For example, a worker with earnings from the predecessor employer in the first quarter and earnings from the successor employer in the second quarter does not initially demonstrate single-employer continuity of earnings across the quarter "seam" that is necessary to qualify as a Beginning-of-Quarter job, but the successor-predecessor edit credits that worker with being employed. Typically, the successor-predecessor edit adds about 1% of jobs that had earnings in the previous or current quarter at separate employers. The successor-predecessor edits were first applied to the LODES data in the 2011 data year (in LODES 7). Earlier years of LODES data do not have this edit.

Establishment Characteristics Edits

In older iterations of the LODES processing system, it was necessary to complete missing establishment characteristics that resulted from the unit-to-worker (U2W) imputation mentioned in Section 4.2.2. However, now these missing characteristics are addressed by using the same methodology as in the main LEHD data processing systems, specifically using data provided by the `u2w_[ST]_supplement` file.[29] See Vilhuber and McKinney [2014, Chapter 5] for additional information on the Employer Characteristics File (ECF).

Record Edits

Finally, records are removed from the LODES data process if they meet any of the following conditions:

[28]While all successor-predecessor edits in the LEHD processing system use similar methods, future work will focus on having all processes utilize a common code base for this edit. The LODES system makes use of a version of the Personal History Enhanced Across SEIN And NONSEIN Transitions (PHEASANT) code base.

[29]These changes were implemented with LODES Version 7.4, which was released on August 29, 2019: `https://lehd.ces.census.gov/announcements.html#082919`.

Incomplete employer characteristics (other than geography) or demographic characteristics This edit is a failsafe as these characteristics should have been completed in the LEHD data infrastructure processing (as discussed in the previous section). In 2018, exactly zero cases required this edit.

Out of age range Records must have an ICF-sourced age of 14 to 99 during the reference quarter, which is consistent with QWI. Typically this excludes less than 0.1% of jobs. In many states, minors under age 14 may not be employed.

Weights are zero The WHATA applies output weights from the QWI production process that can be used to normalize county employment totals from QWI to QCEW totals. These weights incorporate a workplace protection factor (see Section 4.3.2), as well as a factor resulting from the U2W imputation of workers to establishments. The existence of these multiple factors can cause the overall weight for an observed job to be 0. If the final weight of a record is 0, then the record is removed from the data to save file size and processing time.", and a factor resulting from the U2W multiple imputation step. If the final weight of a record is 0, then the record is removed from the data to save file size and processing time. In the latest year of data 0.3% of records were removed with this edit.

4.2.2 Imputation Procedures

Within the processing of LODES, we perform a number of workplace and residential imputations for workers that are missing workplace or residential information. While LODES data processing uses imputation models from LEHD data infrastructure processing to complete job and worker characteristics, LODES requires greater precision for the workplace and residence geography than those general-purpose imputation models produce on their own. This section describes each of these procedures, starting with the unit-to-worker imputation, and then describing the workplace and residence imputations.

Establishment (Unit) to Worker Imputation

While not strictly part of the LODES processing as defined in Section 4.1, the unit-to-worker (U2W) imputation has important effects on LODES and is described briefly here. Approximately 44% of LEHD jobs are at employers with multiple establishments (those filing a Multiple Worksite Report[30]) and, except for in Minnesota, employers do not report to which establishment workers are assigned. The LEHD program uses an imputation model with parameters based on the Minnesota data to draw establishments for workers at multi-unit employers. The U2W imputation model attempts to replicate the reported size distribution of workers across establishments and the observed distribution of commute distances by placing workers at establishments spanning their employment spell that are larger and closer to home. Additional details about the U2W process can be found in Abowd et al. [2006] and Green et al. [2017], with updated analysis in Tucker et al. [2024]

Workplace Imputations

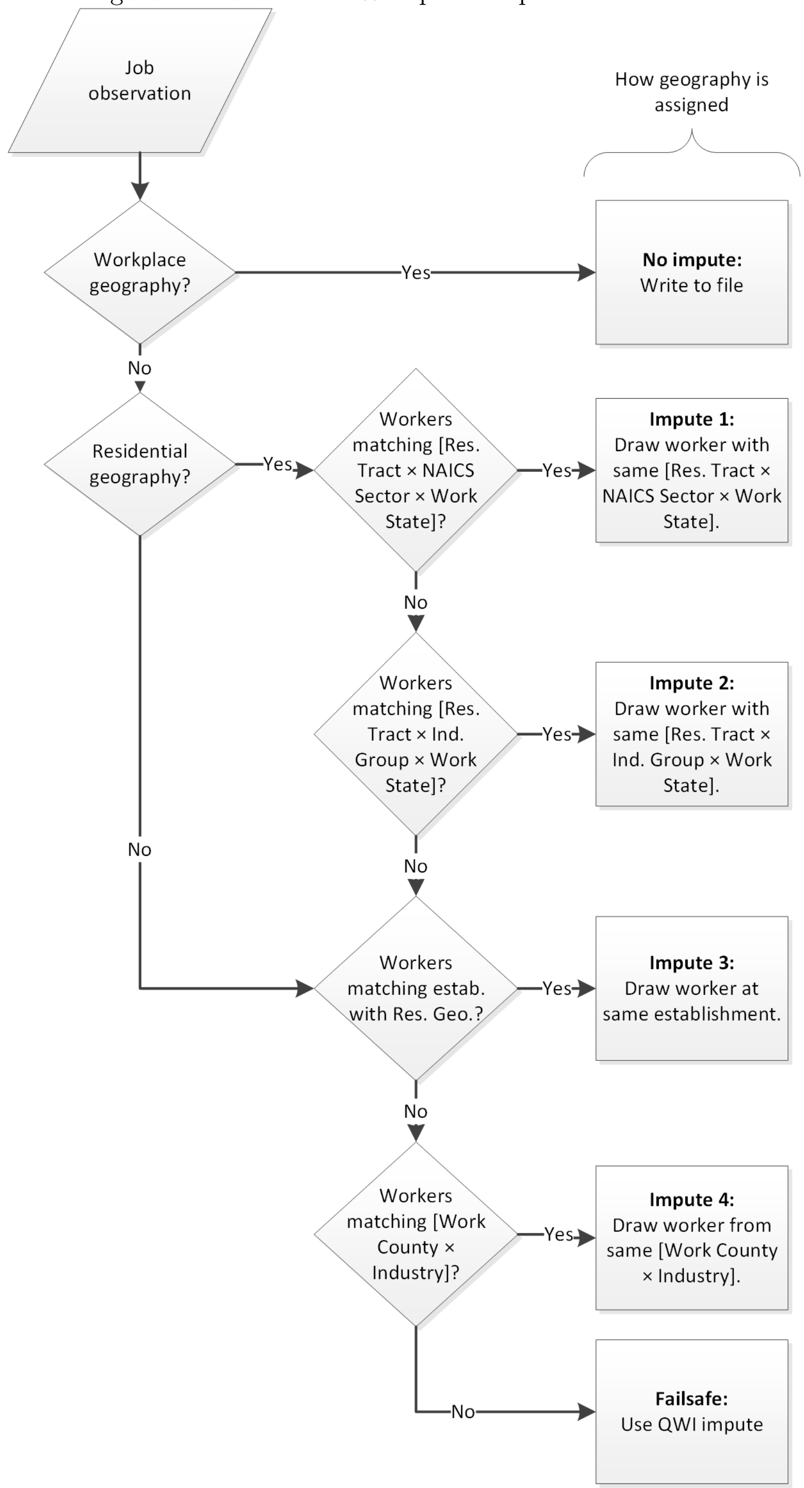
The USDOM process assigns a workplace imputation flag when complete workplace geography (census block) is unavailable. If a workplace has a poor geographic quality, the process assigns the imputation flag one of four possible values, which correspond with the imputation used to complete the geography. The imputation used depends on the available information for the observation (e.g. whether a valid residence location is available). These imputations are hierarchical such that if an observation fails the first imputation, it is passed to the second, etc. A failsafe assignment is applied for any observations that fail the fourth imputation. A flowchart of imputations is displayed in Figure 2.

First workplace imputation For each job assigned to the first workplace imputation, we randomly sample (with replacement) into the distribution of jobs with non-missing workplace geography from workers who live in the same Census tract, work in the same NAICS sector, and work in the same state, and then assign that sampled job's workplace location (census block) to the observation with missing geography.

The first workplace imputation can fail in two ways: 1) The worker has no residence geography, or 2) There is no support (other jobs) in the cell defined by [residence tract × NAICS sector × workplace state]. If the worker has no residence geography, then the observation is passed to the third workplace imputation; otherwise the observation is passed to the second workplace imputation.

[30]For more information on Multiple Worksite Reports, see `hhttps://www.bls.gov/respondents/mwr/home.htm` .

Figure 2: Flowchart of Workplace Imputation Process



Second workplace imputation For each job assigned to the second workplace imputation, we randomly sample (with replacement) into the distribution of jobs with non-missing workplace geography from workers who live in the same Census tract, work in the same industry segment,[31] and work in the same state, and then assign that sampled job's workplace location (census block) to the observation with missing geography.

The second workplace imputation can fail in two ways: 1) The worker has no residence geography, or 2) There is no support (other jobs) in the cell defined by [residence tract × industry segment × workplace state]. In either case, the observation is passed to the third workplace imputation.

Third workplace imputation For each job assigned to the third workplace imputation, we randomly sample (with replacement) into the distribution of jobs with non-missing workplace geography held by workers who are employed at the same establishment, and then assign that sampled job's workplace location (census block) to the observation with missing geography. (Given two jobs at an establishment that had missing workplace geography, if only one received an imputation from the first or second step, then in the third step the job that still has missing geography will be assigned the same workplace location as the job that received a workplace location in the first or second imputation.)

The third workplace imputation can fail in two ways: 1) There is no support at the establishment (no other jobs at the establishment), or 2) None of the jobs at the establishment have an acceptable workplace geography. In either case, the observation is passed to the fourth workplace imputation.

Fourth workplace imputation For each job assigned to the fourth workplace imputation, we randomly sample (with replacement) into the distribution of jobs with the same work county and industry, and then assign that sampled job's workplace location (census block) to the observation with missing geography.

The fourth workplace imputation can fail in two ways: 1) There is no support in the cell defined by [industry × workplace county] (no other jobs in that cell), or 2) None of the jobs in the cell have acceptable workplace geography. In either case, the observation is passed to the failsafe workplace assignment.

Failsafe For each job assigned to the failsafe assignments, we assign the workplace to the initial geocode assigned in the ECF. This geocode is usually of poor quality, and is often at the center of the county, irrespective of the distribution of employment in that county. This assignment happens very rarely.

Residential Imputations

Similar to the workplace imputations, the USDOM assigns an imputation flag for observations that have missing or poor quality residential locations. Once again, the imputations are hierarchical, such that if an observation fails the first imputation, it is passed to the second, etc. A failsafe assignment is applied for any observations that fail the third imputation. A flowchart of imputations is displayed in Figure 3.

First residential imputation For each worker assigned to the first residence imputation, we randomly sample (with replacement) into the distribution of workers who share the same residence county, work tract, and earnings category, and then assign that sampled worker's residence geography (census block) to the observation with missing geography.

The first imputation can fail in two ways: 1) The worker has no county residence geography, or 2) There is no support (other workers) in the cell defined by [residence county × workplace tract × earnings] category. In either case, the observation is passed to the second residence imputation. Note that the support in this case must come from other workers with *non-imputed* workplace geography.

Second residential imputation For each worker assigned to the second residence imputation, we randomly sample (with replacement) into the distribution of workers who work in the same tract and are in the same earnings category, and then assign that sampled worker's residence geography (census block) to the observation with missing geography.

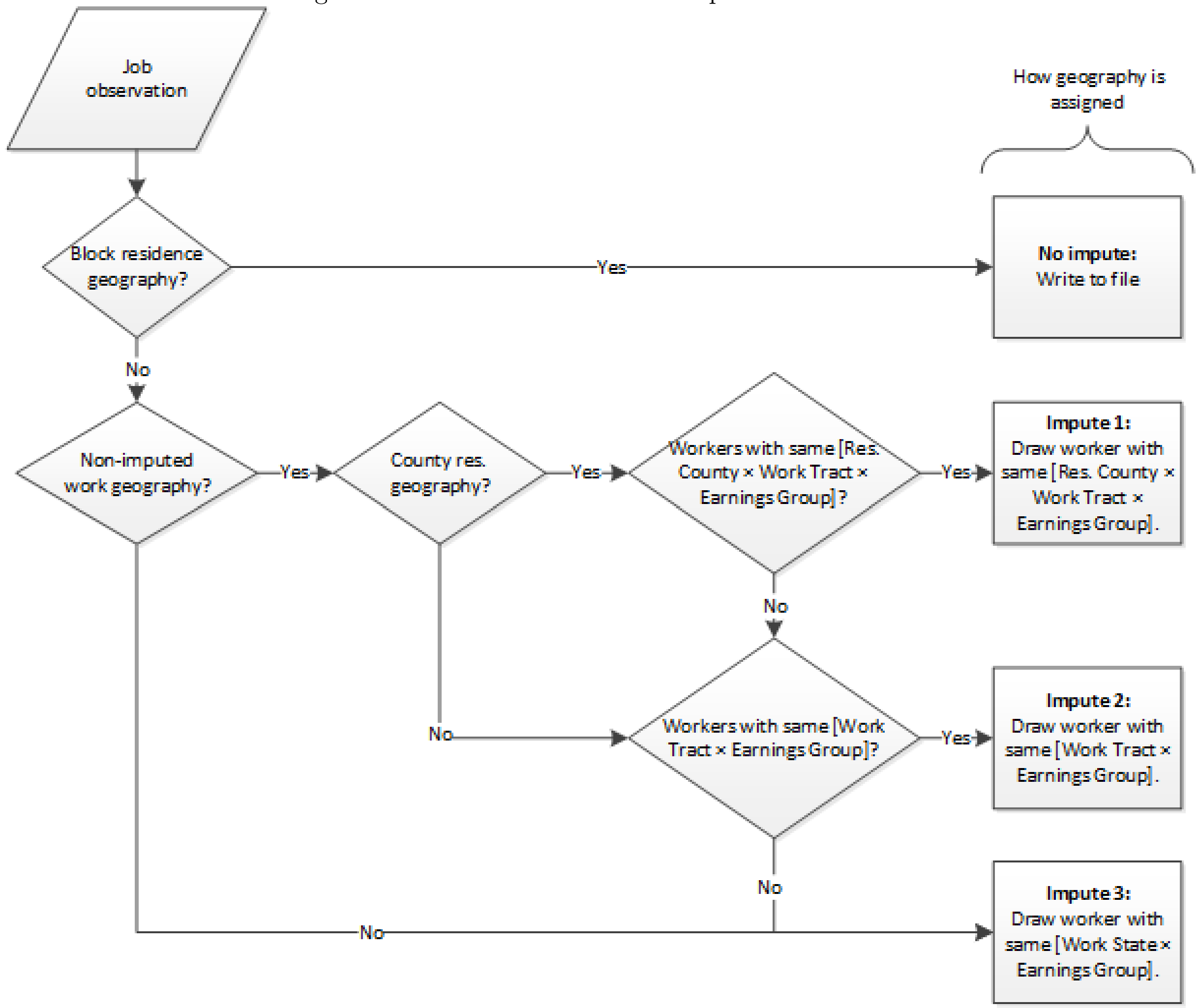
The second imputation can fail in one way: There is no support (other workers) in the cell defined by [workplace tract × earnings] category. If this is the case, then the observation is passed to the third residence imputation.

Third residential imputation For each worker assigned to the third residence imputation, we randomly sample (with replacement) into the distribution of workers who work in the same state and are in the same earnings category, and then assign that sampled worker's residence geography (census block) to the observation with missing geography.

There are no failures in this final imputation because all *residence state* × *earnings category* cells have support with workers who have fully detailed residence geography.

[31] Industry segments are broad categories used in LODES, "Goods Producing," "Trade, Transportation and Utilities," and "All Other Services," the definitions of which can be found in U.S. Census Bureau [2020b] and in Table 10 of this paper

Figure 3: Flowchart of Residence Imputation Process



4.3 Disclosure Avoidance Methods

LODES uses several methods to prevent disclosure of confidential data, including dynamically consistent noise infusion, partially synthetic data, and, for some elements, synthetic data generated with an algorithm that provides probabilistic differential privacy (technical terms described below). The methodologies are approved for use in LODES by the Disclosure Review Board at the Census Bureau. The methodologies are chosen for each element of the data both to provide acceptable data quality and to adhere to the applicable privacy rules. While methodologies implemented in LODES have been at the forefront of research on disclosure avoidance, the Census Bureau continues to consider and develop alternate methodologies that might further improve data quality or the scope of publishable data and maintain confidentiality.[32] Some of the procedures described here, most specifically the assignment of secondary characteristics and residential geography decoarsening, are not explicitly disclosure avoidance procedures. We include discussion of them here as they are essential for making the protected data optimal for public use. While the methodologies presented here have been disclosed certain parameter values are confidential and are not disclosed in this document.

[32] See Haney et al. [2016] and Haney et al. [2017].

4.3.1 Policy Rules Governing Disclosure Avoidance

The use of disclosure avoidance in LODES, QWI, and all Census Bureau data products is mandated by law. Specifically, U.S. Code Title 13 Sections 8b and 9 require that the Census Bureau must not "make any publication whereby the data furnished by any particular establishment or individual under this title can be identified" but may "furnish copies of tabulations and other statistical materials which do not disclose the information reported by, or on behalf of, any particular respondent, ..."[33] In keeping with this mandate, LEHD data products must protect the identities of both establishments (or employers) and individuals.[34] While Title 13 governs protections for data on establishments and individuals collected by the Census Bureau, the Internal Revenue Service (IRS) places additional restrictions on the data that it provides to the Census Bureau; these are codified in Title 26 of the U.S. Code.[35]

The Disclosure Review Board has interpreted the disclosure avoidance mandate to apply differently to establishments and individuals. While the existence of a job and the individual holding it is confidential, the number of employer businesses within a geography and type (e.g. industry, ownership) is not protected by additional disclosure avoidance procedures beyond de-identification of the entity except when required by other data-supplying agencies. Even so, data on the operations of a business (e.g. exact employment count) must be protected. As an example of how this rule applies in other contexts, the County Business Patterns, a dataset derived from the Business Register, the Economic Census, and the annual Company Organization Survey, may publish the exact count of establishments in a location by industry type, even if there is only one such establishment, but may not publish the exact employment count or payroll at such an establishment.[36] In the context of LEHD data products, these rules have led to innovations in protecting the longitudinal series of workplace characteristics, an aggregate of one or more establishments of a type at a location. For LODES specifically, the rules necessitated the development and implementation of synthetic data methods satisfying probabilistic differential privacy to protect the residential locations associated with workers' jobs.

4.3.2 Workplace Job Counts

LEHD applies several disclosure avoidance measures in generating statistics on employment and earnings at workplaces for QWI and LODES. These methods, described in Abowd et al. [2012] and McKinney et al. [2020], avoid any exact disclosures for establishments or workers and distort values by infusing noise at the employer and establishment levels and by weighting to control totals. Noise infusion is designed to be "dynamically consistent," meaning that the same noise factor is applied to a worksite over time, so that longitudinal changes in employment are not distorted. As a short hand, these noise infusion multipliers are known as "fuzz factors." In the construction of QWI tables, cells that are deemed too small are suppressed. In LODES, a different approach is used for small cells, as described below.

Figure 4 illustrates the main steps in producing the protected, or fuzzed, workplace margin of job counts. These protected margin counts serve two purposes: as a direct input to public-use tabulations and also as an input to the protection of secondary characteristics and residence flows.

Primary Characteristics

Ownership, industry segment, earnings, age, sex, race, and ethnicity. These characteristics are selected to have analytical validity, either alone or in combination, for all workplace, residence, and origin-destination tabulations. The data development team selected these characteristics to be representative of the main inputs to LODES, including information on employers, jobs, and workers.

The first step is to aggregate the confidential jobs data by workplace geography and primary characteristics. Workplace tabulations are by census block, as this is the most granular geographic margin released in LODES. Each tabulation cell aggregates both the confidential count of jobs, termed "unfuzzed," as well as count with QWI noise infusion and weighting factors applied ("fuzzed" counts).

Jobs derived from OPM source data, on a subset of federal workers, are not "fuzzed," though they are weighted to match control totals. The data provider already releases tabulations of these jobs to the public and further noise

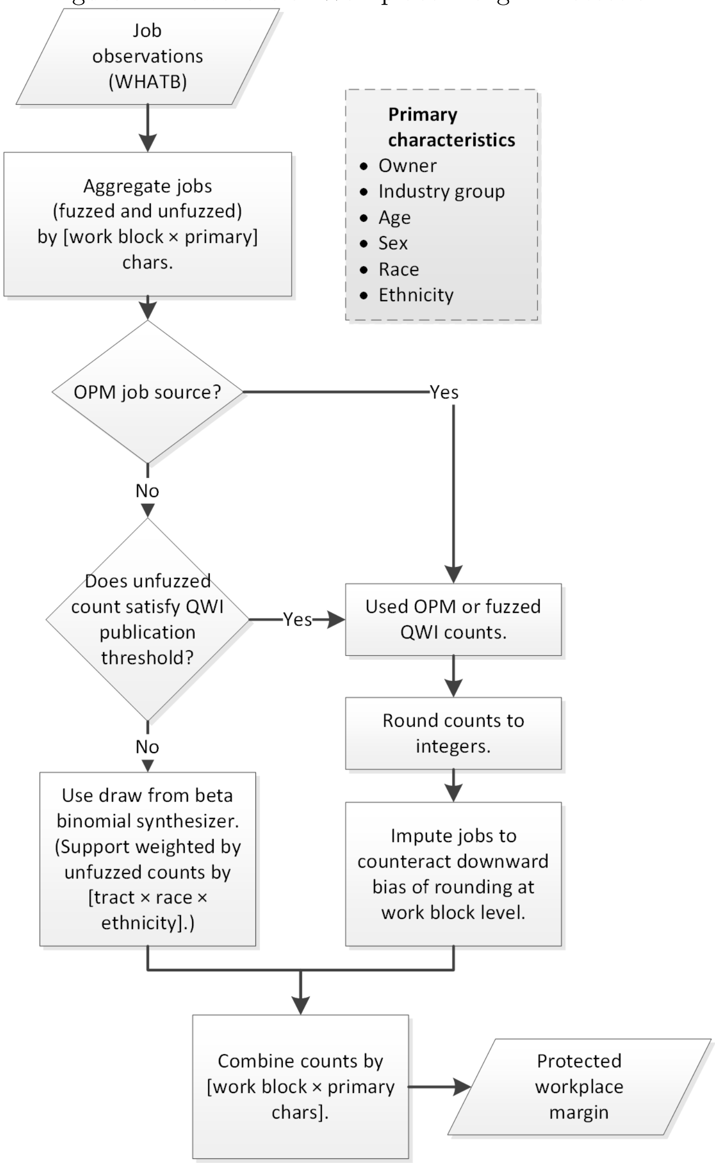
[33] For a review of data protection and privacy at the Census Bureau, see: `http://www.census.gov/privacy/`. For information on Title 13 and the Census Bureau, see: `https://www.census.gov/about/history/bureau-history/agency-history-timeline/title-13.html`

[34] See discussion in Haney et al. [2016] and Haney et al. [2017].

[35] See `https://www.census.gov/about/history/bureau-history/agency-history-timeline/title-26.html`.

[36] See `https://www.census.gov/programs-surveys/cbp/technical-documentation/methodology.html` and Evans et al. [1996] for more information.

Figure 4: Flowchart of Workplace Margin Protection



infusion would not enhance disclosure avoidance.[37] These aggregations go straight to the rounding step.

For jobs sourced from state partners, cell-level confidential aggregations are evaluated for QWI suppression rules and then protected in one of two ways. For those cells with rounded, "unfuzzed" counts satisfying the QWI rule, the "fuzzed" aggregations, along with the OPM data, are rounded to the nearest integer. LODES only reports integer job counts. Because the rounding produces some downward bias on counts, an imputation step adds back in jobs at the workplace block level with probabilities proportional to characteristics at that level. Note that a modest upward bias remains in published totals, which could be addressed in future enhancements.

For the remaining cells where QWI suppression rules apply, a small cell synthesizer draws a count of jobs. For these cells, a Beta Binomial distribution with a support weighted by the confidential count of jobs in the same workplace tract and by race and ethnicity is used to generate a Cumulative Density Function (CDF) across the support. Given

[37]For additional discussion of OPM data, see also: Vilhuber [2018, sec. 6], `https://lehd.ces.census.gov/doc/PressRelease202012Q2beta1.pdf`, `https://lehd.ces.census.gov/doc/help/onthemap/FederalEmploymentInOnTheMap.pdf`, and `https://lehd.ces.census.gov/doc/help/onthemap/LODESDataNote-FedEmp2015.pdf`.

the high degree of interaction among the primary characteristics, small cells are common and application of the small cell synthesizer is widespread.

Synthesizer/Imputation

*Note that a **synthesizer** is, in principle, no different from an **imputation** model. LODES processing generally uses the term imputer for data completion or implementing models, while synthesizer is reserved for disclosure avoidance models. Both models draw records from a CDF defined over support cells with probabilities covering the interval from zero to one.*

Draws from the small cell synthesizer are then combined with the rounded counts and together form the protected workplace margin.

4.3.3 Residence

Residence locations are especially sensitive, as they are both a worker characteristic and derived from federal administrative records including tax return information.[38] Machanavajjhala et al. [2008] describe the design theory for place of residence protection methodology used here. The randomization algorithm satisfies probabilistic differential privacy (see definition below) and was customized for use in LODES.

Figure 5 illustrates the main steps in producing the residence flow margins of job counts. These counts are an intermediate step in generating public use tabulation of residence flows and are also an input to the protection of secondary characteristics. The CDFs used by the residence synthesizer are generated in the SYNTHPREP code process, while the actual synthesis occurs in the SYNTHESIZE process.

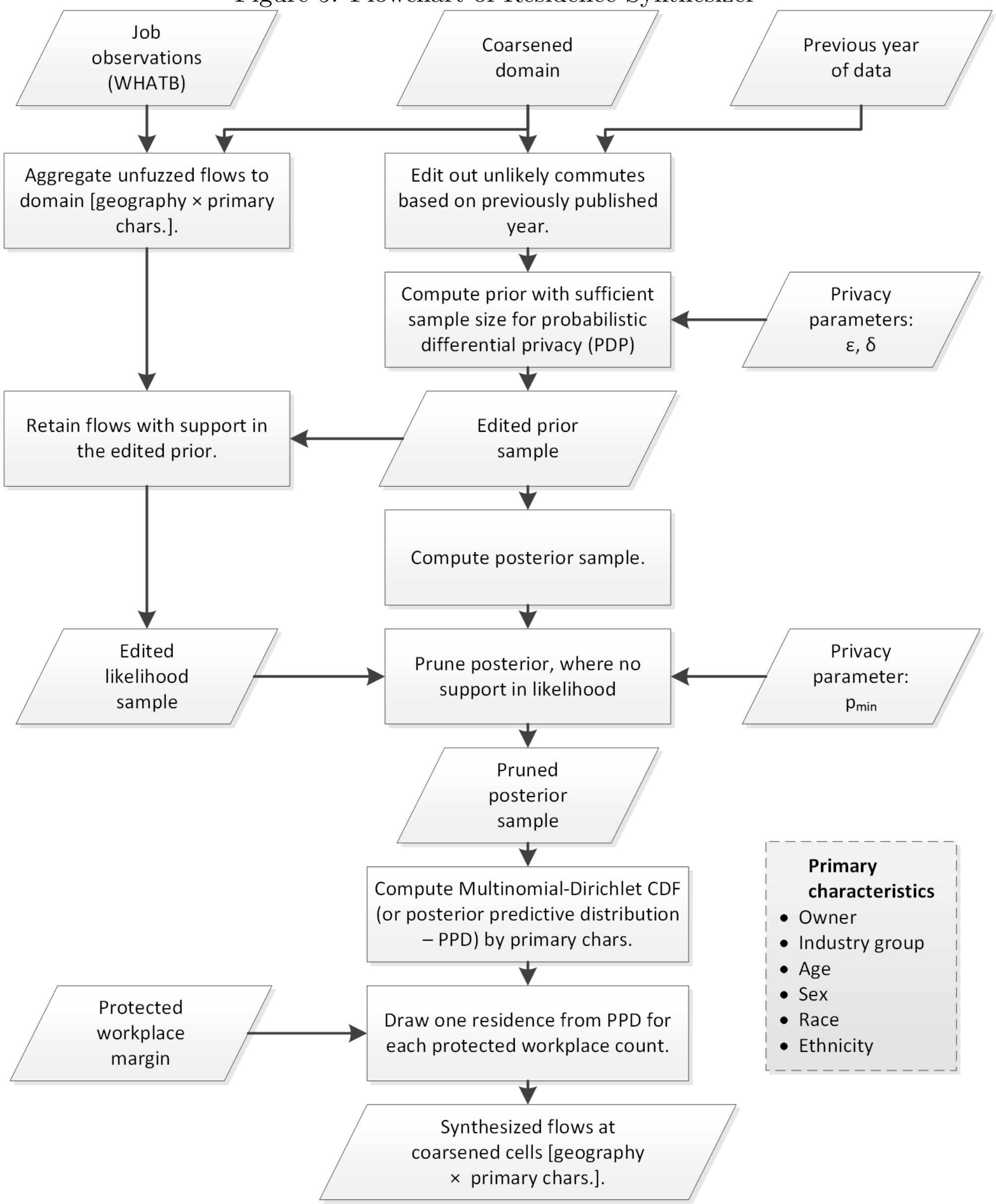
Why LODES uses Probabilistic Differential Privacy

*Machanavajjhala et al. [2008] build on the concept of **Differential Privacy** from the disclosure avoidance literature. Suppose a private database consisting of indivduals (rows) and characteristics of the individuals (columns). A disclosure avoidance algorithm that satisfies differential privacy provides minimal information about any row in the private data. Effectively, it should not be possible for someone using the released data to distinguish whether an individual contributed to the private input data. With differential privacy, the extent of information disclosed about any individual is quantified by privacy loss parameter, $\epsilon \geq 0$ [Dwork, 2006]. A greater value of ϵ indicates more privacy loss, but generally corresponds to improved accuracy of the released data. The accumulation of privacy loss in a system and the policy limit on that value are sometimes referred to as a "privacy loss budget." A challenge in developing a privacy algorithm for LODES was that the residence for any job must be synthesized across a large domain, the entire country. Consequently, the amount of noise required to satisfy the differential priacy criteria would swamp much of the signal and the data would not be useful. The algorithm developed for LODES uses **Probabilistic Differential Privacy (PDP)**, which maintains high data quality while still bounding the worst case of disclosure. With PDP, the probability of a significant disclosure about an individual is at most $0 < \delta < 1$, with δ set to be a very small value. Because the privacy guarantee in LODES is for an individual, data quality on residence flows improves as more jobs are included in a query. For example, a query of the number of jobs with an origin in one (large) county and a destination in another (large) county will be less noisy than a query of the number of jobs with an origin in one census tract and a destination in another census tract. Additional work by the Census Bureau on disclosure avoidance can be accessed at `https://www.census.gov/about/policies/privacy/statistical_safeguards.html`.*

The first step defines the complete set of possible flows, labeled the Domain, using the hierarchy of Census Bureau geographic relationships, which is public. In principle, the domain includes flows between all possible geographic pairings. In practice, the possible flows are aggregated using the hierarchy of geographic classifications, or "coarsened," to improve data quality (Machanavajjhala et al. [2008]). The coarsening retains granularity for flows between nearby locations (where commuting is intense and we generally expect a high signal-to-noise ratio), but aggregates the residence location dimension for longer distances (where flows tend to be more sparse). Both workplaces and residences are aggregated, at minimum, to the census tract level. Residences are further aggregated to Public Use Microdata

[38]While some inputs to the RCF are sourced from tax data, the RCF itself has been determined to be only protected under Title 13.

Figure 5: Flowchart of Residence Synthesizer



Area (PUMA) and Super PUMA levels (from 2000) as flows become longer distance. In general, there is no aggregation beyond the tract level within a metropolitan area. Protected flows will be "decoarsened" later, to census blocks, in order to achieve the expected publication detail.

The coarsened domain is used in conjunction with public-use job flows data to further improve data quality by editing out unlikely commutes in the prior and the likelihood. While it is fundamental to PDP that the domain includes flows to all possible residences, even the coarsened flows are still so numerous that including them all would require an excessively diffuse prior distribution and degrade data quality (Machanavajjhala et al. [2008]). LODES uses public-use job flows data from the previous year (or from the 2000 Census Transportation Planning Package in the first year of LODES) to inform an edit, in which these counts are tabulated by the coarsened domain and some especially unlikely (long distance) flows not appearing in the previous year data are randomly removed based on probability shown in Table 7. The remaining job flow counts from the previous year, termed the "edited prior

Table 7: Coarsened Domain Probabilities

Distance Range [miles]	Retention Probability
[0,100)	100%
[100,200)	50%
[200,500)	10%
[500,max]	2%

sample" or just the "edited prior," are then augmented with an algorithm described in (Machanavajjhala et al. [2008]). This algorithm uses the privacy protection parameters ϵ and δ to compute a prior sample with sufficient size for all possible flows from a workplace to ensure PDP.

In parallel, the unfuzzed, or confidential file is aggregated by the coarsened domain of residence flows. Aggregation is by every combination of the primary characteristics. The objective of this process is to protect these counts. The aggregated flows are further restricted by only retaining those counts with a matching cell that was retained in the edited prior, or the edited, coarsened domain. The resulting file may be referred to as the "edited likelihood sample" or "edited likelihood."

The edited likelihood sample and edited prior sample are added using the Multinomial-Dirichlet conjugate relationship to form a posterior sample. To improve data quality, the Dirichlet posterior is "pruned" to remove some flows with no support in the edited likelihood (Machanavajjhala et al. [2008]). The probability that a flow is pruned is the maximum of p_{min}, a third protection parameter, and an inverse function of the number of possible flows, or support points. More extensive pruning (with higher p_{min} or a higher prune probability) improves data quality but results in more privacy loss (still limited by the very small value of the allowable failure rate, δ). The resulting origin-destination job counts are then formed into a CDF, with the edited, coarsened domain as a support and probabilities set by the pruned posterior sample size. This CDF is also know as a Posterior Predictive Distribution.

The final step is to combine the pruned posterior CDF with the protected workplace margin (see 4.3.2) and draw, or synthesize, a single residence location for each job in the protected workplace margin. As the pruned posterior is defined by workplace census tract and primary characteristics, the same CDF applies to all jobs in a workplace tract with the same primary characteristics, even if that tract contains multiple workplace blocks. Each job receives only one draw, and the drawn residence is assigned to that job. This set of draws, known as an implicate, satisfies the definition for PDP.[39]

4.3.4 Secondary Characteristics

Secondary characteristics are supplemental to workplace, residence, and the set of primary characteristics. These are available in tabulations by workplace and residence, but not for origin-destination flows. They are not factored into protection of the workplace job count. Rather they are estimated for the set of protected jobs with models informed by the confidential job counts.

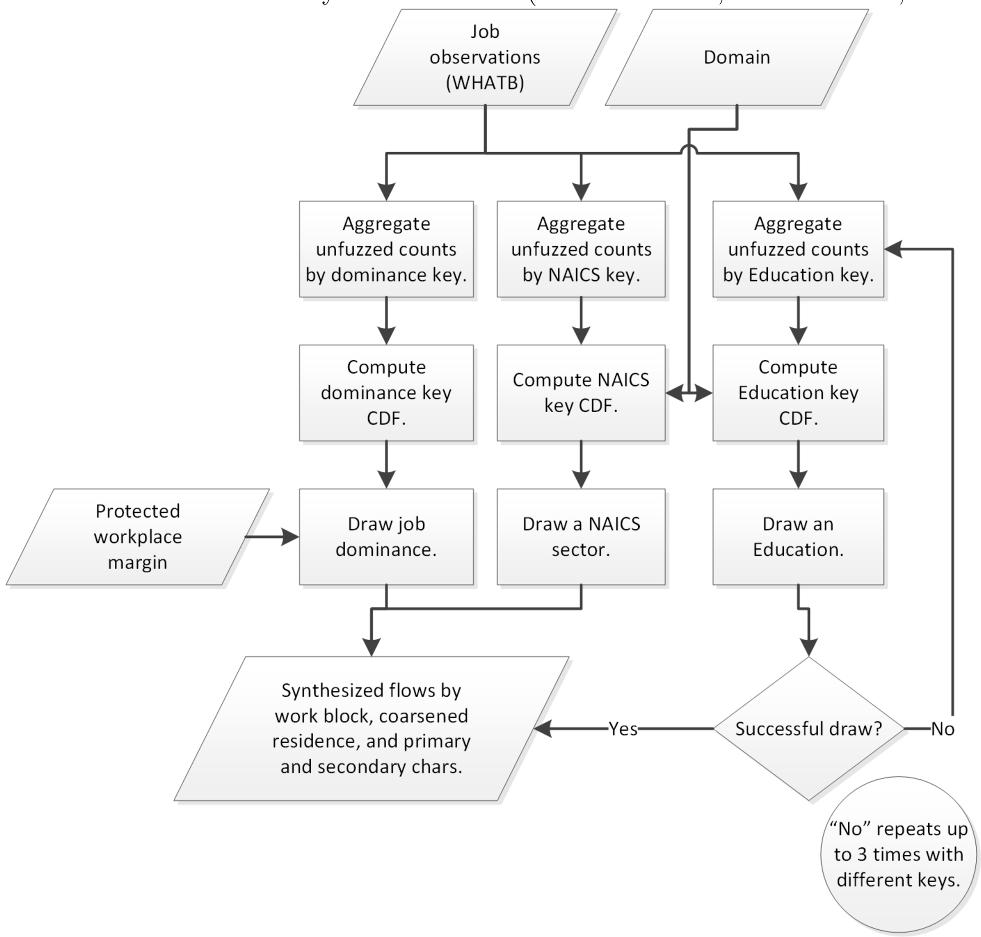
This section describes how secondary characteristics are assigned in LODES. Figure 6 illustrates the main steps in producing the secondary characteristics (Job Dominance, NAICS Sector, and Educational Attainment) for protected jobs (protected workplace margin). Figure 7 illustrates the steps for producing the remaining secondary characteristics, firm age and firm size. In both cases, these counts are an intermediate step in generating public use tabulation of workplace and residence job counts. The CDFs used for secondary characteristics are generated in the SYNTHPREP code process, while the actual synthesis occurs in the SYNTHESIZE process.

Secondary Characteristics

Secondary characteristics include job dominance, NAICS industry sector, educational attainment, and employer firm age and size. A dominant job is the highest earning job for a worker at the beginning of the second quarter, and is released in conjunction with total job counts for all public use tabulations. The twenty NAICS sectors are nested within the three industry groups. Educational attainment includes four categories as well as a null value for persons under age 25. Firm age and size are defined only for private sector employers.

[39] Drawing multiple implicates (and reweighting each implicate by the inverse of the count of implicates) would improve accuracy but degrade disclosure avoidance.

Figure 6: Flowchart of Secondary Characteristics (Job Dominance, NAICS Sector, Education)



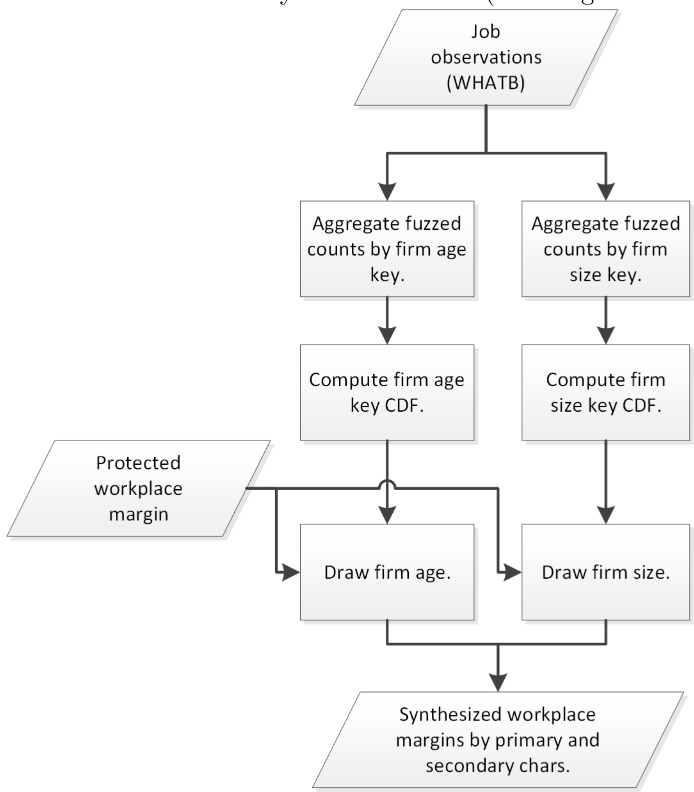
Secondary characteristics, not included in the primary characteristics, must be estimated because all other characteristics, as well as direct identifiers (establishment, person, job), are lost in the protection of workplace job counts. The estimation support for secondary characteristics consists of a separate key, or set of stratifying variables, with some characteristics having multiple, fall-back keys. Job dominance is estimated for a key defined by workplace block and the primary characteristics. NAICS sector uses a key defined by workplace block and the primary characteristics. Educational attainment uses an initial key defined using workplace block, the residence domain coarsened to PUMA, and the primary characteristics, as well as fall-back keys with more aggregated workplace and geography as well as no residence geography. Firm age and size uses a key defined, only for private sector employers, by workplace block. Firm age and size are only published by workplace geography.

For each characteristic, a CDF is constructed for the margin defined by the respective key, with probabilities based on the confidential count of jobs in each characteristic-by-geography cell. Though the procedural steps for each imputation differ, the process may be thought of as follows. The protected workplace margin, along with protected coarsened residence flows from the residence synthesizer, are merged by the key for each support with the appropriate CDF. Each protected job draws a value for each secondary characteristic, as appropriate given the definitions. In the case of a support failure, fall-back keys are used.

4.3.5 Decoarsening

After the steps above, the resulting set of jobs and their workplace, residence, primary characteristics, and augmented secondary characteristics are considered to be protected at this point in processing, but not yet in an appropriate form for public use. To make the job flows data suitable for arbitrary queries by workplace or residence geography,

Figure 7: Flowchart of Secondary Characteristics (Firm Age and Firm Size)



the protected residence locations must be disaggregated, or "decoarsened," to the census block level. This section describes how census blocks are assigned to protected residences in LODES.

Figure 8 illustrates the main steps in this decoarsening procedure. The decoarsened jobs constitute a protected microdata file, which is used for data quality analysis and other research purposes and from which the public-use files are tabulated. The protected microdata file undergoes postprocessing, including the application of geography definitions as well as aggregation along various dimensions, before it is ready for public use. The CDFs used for decoarsening are generated in the SYNTHPREP code process, while the actual imputation occurs in the SYNTHESIZE process.

The distribution of population from the most recent decennial Census of Population and Housing informs the decoarsening model. From Summary File 1, persons are counted by residence block, race, and ethnicity. These totals are aggregated to census tract, PUMA, and Super PUMA (as the tabulation area was labeled in 2000), to match the support of the coarsened domain. At each level of the coarsened domain (Super PUMA, PUMA, tract), a CDF is computed for a support of cells for the next level down (respectively: PUMA, tract, block), with probabilities defined by the count of persons in each support cell.

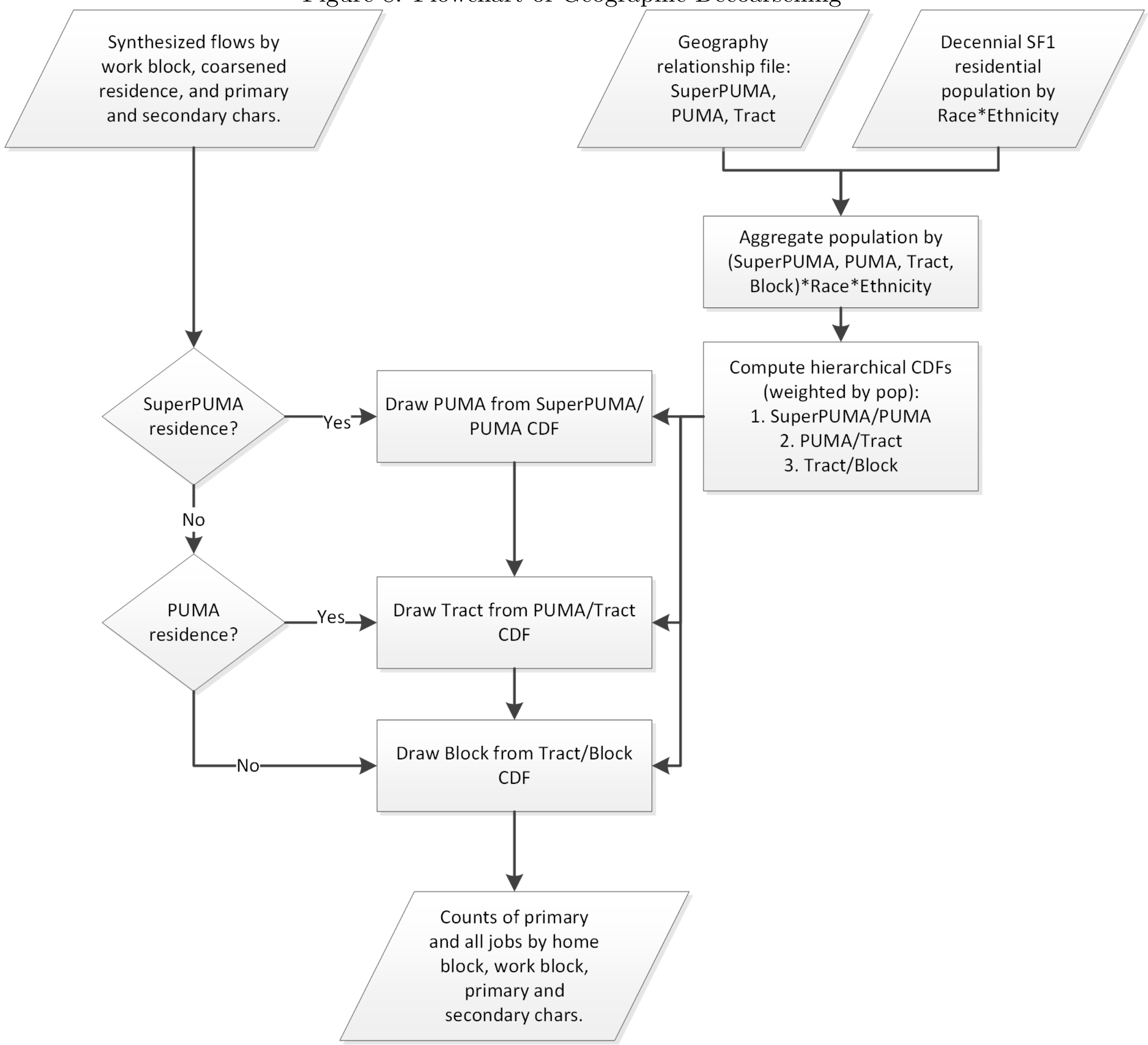
The protected jobs are merged to the CDF with a matching, coarsened residence support. Starting at the Super PUMA level, a more granular residence is drawn, and the process is repeated at each coarsening level until all jobs have a residence census block assigned. This file is then ready for post processing and public use tabulation by workplace, residence, and origin-destination flow.

5 Quality Review

This section describes the procedures we use during the data processing to ensure data quality. Additionally, we describe the quality metrics that we will publish along with the LODES data, so that end users are able to assess the data quality of a particular tabulation.

Beyond quality review that takes place during a regular production cycle, the LEHD program pursues long run

Figure 8: Flowchart of Geographic Decoarsening



efforts to understand and improve data quality. One example of such efforts that is particularly relevant to LODES is the ACS/LEHD data-linking project, known as "Dev10" [Green et al., 2017]. This project investigated differences in commuting statistics between LODES and ACS. The study found that LODES commutes tend to be longer due to differences in the administrative data workplace frame and due to uncertainty in the assignment of workers to establishments at multi-unit employers.

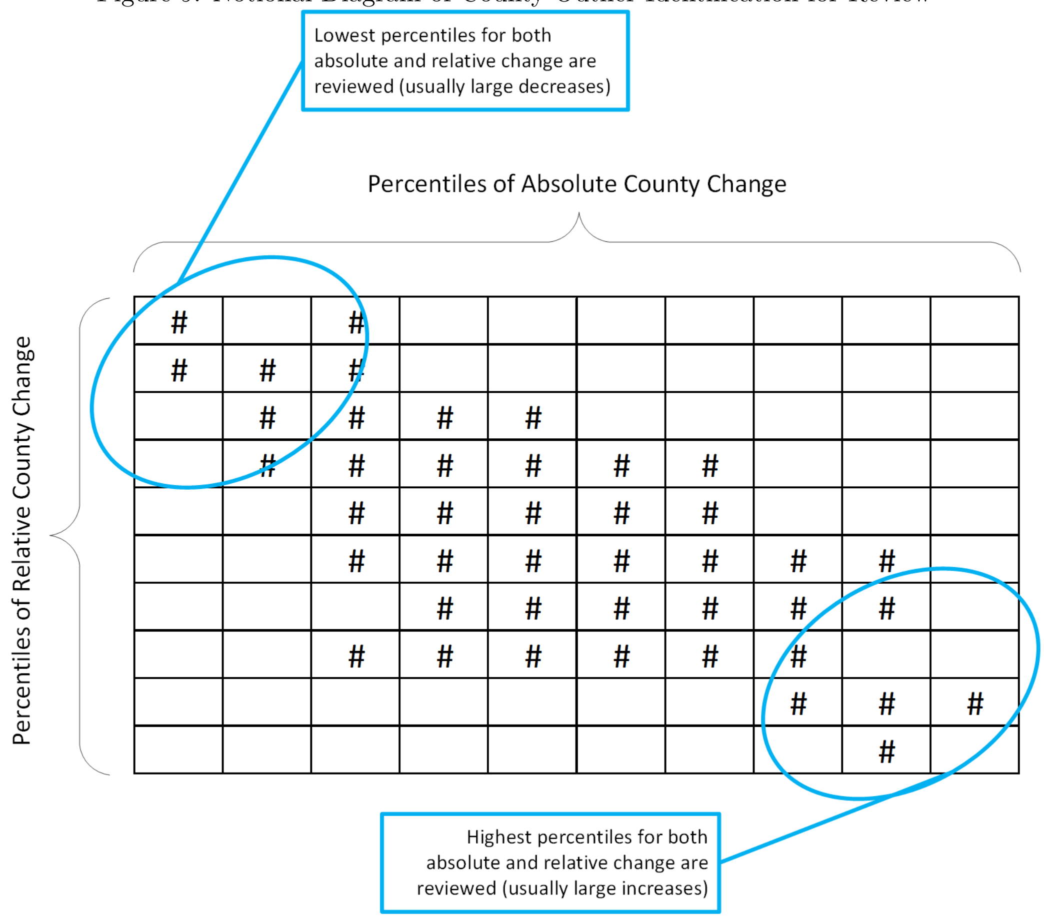
5.1 Quality Review

A number of inline quality assurance checks take place within the LODES data processing, specifically within the USDOM and WHATB processes. Additionally, at the end of the data processing, we have a special module called OTMQA, which does summary checks on the data quality.

In OTMQA, we review the final LODES datasets prior to release at both the work and home margins. The review is performed by comparing the existing public-use, *county-level* LODES data to the current year being processed. The goal of this review is to identify time-series outliers in the most recently processed year of data. Extreme outliers are then investigated by hand to determine if there is an underlying data problem, such as a misreported multiple-establishment firm or an address geocoding problem.

There are two methods that we use to identify these outliers. First, we measure symmetric differences between the data year under review and the previous data year. Second, we compare the data year under review and the distribution of available prior years of data, in both the count and shares. Only those that are at the extremes of both share and count distributions are reviewed. Both of these methods are performed for county data. A notional diagram of the selection of outlier counties for review can be seen in Figure 9.

Figure 9: Notional Diagram of County Outlier Identification for Review



In some cases, this review identifies geocoding problems that mis-located large (relative to total county employment) establishments. When possible, hand edits are made to the geocodes for the miscoded establishments and the relevant portions of the LODES processing system are rerun to correct the public-use totals. In other cases, review identifies what appears to be a "UI hole," in which a firm does not report in a certain quarter; in these cases, there are no edits made to the underlying data. Starting in 2019Q1, the EHF impute was integrated into LEHD production upstream of, which addressed those one-quarter UI holes. Finally, in some rare cases, where geocoding is precise but incorrect and we have definitve evidence of such, we blank the existing geography and force the establishment location to be imputed through the steps above. In most cases when specific issues are identified and can be corrected, these edits are carried forward to future years of processing.

5.2 Quality Statistics

To provide our users with an ability to assess the quality of the geographies with which they are working, we describe a number of quality statistics that we are considering publishing in conjunction with future releases of LODES. These quality statistics fall into three broad categories: workplace data quality, residence data quality, and aggregation convergence accuracy. In the future, other metrics of quality may be published.

Workplace Data Quality

For workplace geography, data quality is affected by five issues:

Nonsampling error Nonsampling error comprises issues such as missing data, as when a firm does not report in the second or third quarters, which lead to those jobs not being counted.

Low quality data inputs Low quality data inputs refer to issues with the data we receive, such as a firm not listing multiple worksites, and instead reporting all employees at a central location. Another example of low quality data inputs is when a firm reports a precise location for an establishment, but that location is not compatible with conventional wisdom. For example, administrative data may say that 1,000 workers are employed at a location, but the location has a very small physical structure and a small parking lot such that only 10 or 20 people could reasonably work at the location. Such cases may not be corrected in our edits and imputations (because we struggle to correctly identify them and/or may not have enough information to correct them), but will still affect the data quality for the user.

Low quality edits/imputations Data quality may also be affected by the quality of our edits and imputations for workplace, which are detailed in Section 5.2. For example, since we use industry segment to impute workplace information, it is possible for a hospital worker, technically in the "All Other Services" industry segment, who lives in the same tract as a retail worker to have their workplace imputed to the nearest mall. This effect is exacerbated if it is imputed in the first step of the imputation process, because it will then affect all other workers at that hospital in the second step. Statistics on workplace imputations are provided in Appendix A in Table 14.

Confidentiality protection Workplace data quality is also affected by the confidentiality protection at the establishment that is used in the QWI infrastructure. More detail on that protection system is described in Section 4.3.

Geocoding Geocoding can be considered a special case of edits and imputations but is identified separately here. Because workplace geocoding at the establishment can shift the location of many jobs, issues with poor geocoding have a larger effect on the quality of workplace counts than on the quality of residential counts, especially at very detailed geographies. Note that providing this information will not help users identify cases where a workplace received precise geography that is not compatible with conventional widsom, or when a large firm reports incorrect employment structure, such as multiple worksites that are unreported. To help users diagnose self-identified data quality issues, we publish tabulations of geocoding quality at the state level. These statistics a provided in Appendix A in Table 16.

It may also be helpful to situate these workplace data quality issues within the context of the components of total variability that are discussed in McKinney et al. [2020, Section 5]. These components are record missingness, disclosure limitation, and characteristic missingness. Our nonsampling error would fall into McKinney et al.'s missing records/employers component; geocoding, low quality data inputs, and low quality edits/imputations would all be covered by their missing characteristics[40]; and the confidentiality protection issues would fall into their disclosure limitation component. Further discussion of comparability to McKinney et al. [2020] appears in Appendix A. Note that residence variability, a feature of LODES, is beyond the scope of the QWI and the analysis of McKinney et al. [2020], but we discuss parts of it in the next section.

Residential Data Quality

Data quality for residence location is affected by three issues:

Nonsampling error Residence information is not available in administrative records for an individual. Additionally, some residential addresses need to be imputed. Statistics on residential imputations are provided in Appendix A in Table 15.

Input data quality Input data quality refers to the quality of the data that the RCF receives. Because the timing of the RCF address and the job timing (April 1) are not necessarily the same, the residence reported may not be the residence that a worker had when working at that job. Additionally, if a worker has multiple addresses, it is possible that the RCF chooses a secondary residence, which could potentially increase measured commuting distance. Statistics on residential imputations are provided in Appendix A in Table 15.

[40] Even though some aspects—worksite missingness—may appear to be record missingness, we still know about the jobs in these cases.

Confidentiality protection Finally, data quality of residence is affected by the confidentiality protection system that is used by LODES. This protection system is described in Section 4.3. Because the released data include noisy synthetic components, they differ from the confidential data.

There are two main ways to summarize how the protection system affects data quality. First, we report tabulations of the true and synthetic counts of workers at the residence block level by state, similar to Abowd et al. [2009] (see Appendix A, Tables 17 and 18). Second, we summarize the data quality of the residence information by measuring the extent to which the true distribution of residences is preserved.

To do the latter, we construct measures of divergence between the confidential and released distributions of residential locations over a polar grid. This grid is defined by 25 cells (distance × direction, see Table 8) with each containing approximately an equal share of jobs in an national summary.[41] These distances and directions are calculated from a representative point within each census tract such that all the residences or workplaces within a census tract use the same point for this calculation.

Table 8: Residential Divergence Grid

Distance Intervals [miles]	Direction Intervals [degrees from North]	
$[0]$[42]	N	$[-22.5, 22.5)$
$(0, 6.8]$	NE	$[22.5, 67.5)$
$(6.8, 18.9]$	E	$[67.5, 112.5)$
$(18.9, max]$	SE	$[112.5, 157.5)$
	S	$[157.5, 180]$ $[-180, -157.5)$
	SW	$[-157.5, -112.5)$
	W	$[-112.5, -67.5)$
	NW	$[-67.5, -22.5)$

For each "workplace margin" of the origin-destination matrix, we calculate two variables for each of these cells: P_{ij}, which is the number of workers in workplace j whose residence tracts are in cell i of the $distance \times direction$ distribution for the confidential data as a share of the the total number of workers employed in workplace j; and Q_{ij}, which is the protected approximation of P_{ij}. Using these two variables, we calculated the Mean Jensen-Shannon Divergence (JSD). The JSD is one of a class of divergence measures that quantifies the difference between a "true" distribution (i.e. the confidential distribution) and an "approximate" distribution (i.e. the released distribution). It also has the convenient properties of being both symmetric, such that $\bar{D}_{JS}(P,Q) = \bar{D}_{JS}(Q,P)$, and bounded [0,1] (when log_2 is used). The JSD is calculated as below:

$$\bar{D}_{JS}(P,Q) = \frac{1}{2}[\frac{1}{N}\sum_{j=1}^{J} n_j \sum_{i=1}^{G} P_{ij} log_2 \frac{P_{ij}}{M_{ij}} + Q_{ij} log_2 \frac{Q_{ij}}{M_{ij}}] \tag{1}$$

Where $M_{ij} = \frac{1}{2}(P_{ij} + Q_{ij})$ and G, the number of grid cells, is 25 as discussed above.

Additionally, two other divergence measures, the Mean Kullback-Leibler Divergence (KLD), and the Root Mean Integrated Square Error (RMISE), are discussed in Appendix A.

We calculate JSD (and the RMISE) for all tracts, as well as by different employment count groups (0-99 jobs, 100-999 jobs, 1,000-9,999 jobs, and 10,000+ jobs), since the divergence can depend upon sparsity/size of the workplace margin. These statistics (JSD and RMISE) are aggregated to the national and state level and shown in Tables 19 and 20 in Appendix A.

Aggregation Convergence

Another useful metric to assess data quality overall is the degree to which public data converges to confidential data with geographic or characteristic aggregation. Some users query LODES for case studies that are very sensitive

[41] Note that this grid is similar in construction tothe Distance/Direction query for a workplace in OnTheMap, though the cell definitoins are not identical.

[42] When distance is exactly 0, the direction is undefined. This case constitutes a 25^{th} category when distance and direction are crossed.

to employment levels in individual census blocks. These analyses can, occassionally, be subject to large anomalies. Within the workplace margins, anomalies may be due to the incorrect, imprecise, or updated geocoding of a large employer. Within the residence margins, anomalies may be due to differences in coarsening of the domain and decoarsening finer geography from year to year, or due to concentrated deficiencies in the residential data.[43] Sizes of many of these anomalies become relatively smaller for a query over a larger geographic area. Although we cannot communicate whether a specific data point has an anomalous value, we could communicate the frequency and severity of anomalies at various levels of aggregation, and for different geographic regions or states. We calculate these quality tabulations at the block, block group, tract and county level. Details of these tabulations are currently maintained internally but may be available for release at a later date.

6 Dissemination

The public-use data files are released by two primary methods: the OnTheMap web application and in "raw" public-use files that can be downloaded from the LEHD website.

6.1 Scheduling

The production of the full LODES data product requires inputs from several key data sources. Each of these has its particular availability and processing lag.

State job/firm data LEHD receives data inputs from states three quarters after they are collected. Processing of the LEHD infrastructure data takes approximately one quarter (although the QWI data for individual states are released on a flow basis), so that a full set of public-use QWI data is typically finished being released with approximately a one year lag. Internal files for LODES are available with approximately the same lag.

Federal job/firm data LEHD receives data from the Office of Personnel Management (OPM) on a quarterly basis. These data are currently processed on an *ad hoc* cycle, but at minimum once per year in advance of the LODES processing. OPM records are processed to resemble the structure of the UI wage records in order to be combined with QCEW records from the states on Federal "firms" and "establishments" so that we can construct a frame of federal jobs very similar to the one constructed for private and local/state employment.

Residence data LEHD receives inputs for the RCF process on an annual cycle. Files for each data year arrive in August/September of the following year. RCF processing takes about one month of processing and QA so that a completed RCF for a data year is available in late autumn of the next year.

The LODES system further processes the outputs of the LEHD infrastructure and typically requires 1-2 months of computing, QA, and file preparation. The reference period for LODES is Q2 of each year (March-June) and so Q2 infrastructure processing must be complete before processing of LODES can begin. For quality reasons, processing of LODES data does not begin until Q3 infrastructure data are available.[44]

As such, the earliest that LODES processing might begin is the beginning of Q4 (October), and the end of Q4 (December) is usually set as the default release date. By the time of this release, the data are lagged approximately 6 quarters from the reference period. Delays in any of the inputs or significant quality problems that arise with any of the inputs can delay processing of LODES and push back the release of the public-use files.

6.2 Published Characteristics and Interactions

Table 9 lists the job-worker-firm characteristics that are part of the pubic-use LODES datasets.

The official LODES public-use data aggregate over several dimensions of the synthesized origin-destination matrix. The collapsed OD matrix has the structure shown below and counts are released for each unique cell. Cells that do not appear in the output datasets have total counts of zero. Missingness is not a concept currently used in LODES.[46]

[43] For example, a number of workers may in either the underlying data or synthetic data be misplaced into a specific block, although in the other dataset they reside in adjacent blocks. In this case, the employment of the block group would be correct, since the errors would average out, but in the specific blocks the employment would be over- or under-counted.

[44] State partners regularly send in corrections to data and waiting until these revisions are included in the following quarter generally improves data quality.

[45] See Table 10 for definitions of the Industry Segments.

[46] A more detailed description of the public-use files can be found in U.S. Census Bureau [2020b].

Table 9: Published Characteristics

Job/Worker/Firm Characteristics	Notes
Work Block	11.1 million possibilities
Home Block	11.1 million possibilities
Ownership	Combined with Dominance into Job Type (see Table 11)
Dominance	Combined with Ownership into Job Type (see Table 11)
Year	2002 through Current
Age	3 categories
Earnings	3 categories
Industry Segment	3 categories[45]
NAICS Sector	20 categories
Race	5 categories
Ethnicity	2 categories
Educational Attainment	4 categories
Sex	2 categories
Firm Age	5 categories
Firm Size	5 categories
See U.S. Census Bureau [2020b] for more details on the published characteristics.	

Table 10: Industry Segment Defintions

Industry Segment	NAICS Sector
Goods Producing	11: Agriculture, Forestry, Fishing and Hunting
	21: Mining, Quarrying, and Oil and Gas Extraction
	23: Construction
	31-33: Manufacturing
Trade, Transportation, and Utilities	22: Utilities
	42: Wholesale Trade
	44-45: Retail Trade
	48-49: Transportation and Warehousing
All Other Services	51: Information
	52: Finance and Insurance
	53: Real Estate and Rental and Leasing
	54: Professional, Scientific, and Technical Services
	55: Management of Companies and Enterprises
	56: Administrative and Support and Waste Management and Remediation Services
	61: Educational Services
	62: Health Care and Social Assistance
	71: Arts, Entertainment, and Recreation
	72: Accommodation and Food Services
	81: Other Services [except Public Administration]
	92: Public Administration

Table 11: Job Type Definition

Job Type	Ownership	Dominance
All Jobs	All	All
Primary Jobs	All	Dominant only
All Private Jobs	Private	All
Private Primary Jobs	Private	Dominant only
All Federal Jobs	Federal	All
Federal Primary Jobs	Federal	Dominant only

$$Work\ Block \times Home\ Block \times Ownership[3] \times Dominance[2] \times Year[17] \times$$
$$(Total[1]|Age[3]|Earnings[3]|Industry\ Segment[3])$$

Additionally, the ownership and dominance categories have been joined into the concept of "Job Type." The makeup of the Job Type variable is given in Table 11.

Public ownership, all jobs without Federal employees, and non-dominant job totals can be calculated via subtraction of the various job types. Additionally, home and work margins are published. They contain additional characteristic detail and the cells are constructed as follows.

$$(Work\ Block|Home\ Block) \times Ownership[3] \times Dominance[2] \times Year[17] \times$$
$$(Total[1]|Age[3]|Earnings[3]|Industry\ Segment[3]) \times$$
$$(Total[1]|Age[3]|Earnings[3]|NAICS\ Sector[20]|Race[5]|Ethnicity[2]|$$
$$Educational\ Attainment[4]|Sex[2]|Firm\ Age[5]|Firm\ Size[5])^{47}$$

6.3 Coverage and Variable Availability

The universe and variable availability of the LODES/OnTheMap data has changed significantly over the lifetime of the data product and are determined by the available sources for the individual wage records as well as other sources for person/business characteristics. Most wage reocrds come from the state partners' Unemployment Insurance systems and this source has formed the bulk of the universe of jobs covered in LODES/OnTheMap. A complimentary source of wage records is the Office of Personnel Management (OPM) through which data on Federal jobs are made available.

Since the inception of LODES/OnTheMap, the number of states whose data have been included has changed. With the release of LODES 7.5[48], 49 states and the District of Columbia are available for 2018. Some of the states, though, do not have complete data histories due to missing data history or currently expired data sharing agreements. The coverage and variable availability and year are given in Table 12.

As Table 12 shows, data on Federal employment are currently available for data years 2010 through 2018. Additionally, it is possible for one or more states (or other data source) to be held from future publication based upon quality reviews. In these cases, the coverage would contract. When possible, these data quality issues are corrected and missing state-year combinations are restored to the public-use data.

6.4 Public-Use Data Files

Public-use LODES datasets are currently released for download on the LEHD website at `http://lehd.ces.census.gov/data/#lodes`. Technical documentation for users wishing to work with the raw files is available in U.S. Census Bureau [2020b]. The full file structure is described in the technical document but an overview is given here.

LODES data files are separated by state of employment for each job. Within each state are origin-destination (OD), residence area characteristics (RAC), and workplace area characteristics (WAC) files that are separated by year,

[48]See `https://lehd.ces.census.gov/announcements.html#120820` for more details.

[49]While Puerto Rico and the U.S. Virgin Islands have been part of the LED partnership, no LODES data was ever produced for them.

[50]Firm age and firm size variables are only available for "All Private" jobs.

Table 12: LODES Coverage and Variable Availability

Year(s)	Count of Available States	Unavailable States[49]	Federal Jobs	Ownership, Dominance, Age, Earnings, Industry variables	Race, Ethnicity, Education, Sex variables	Firm Age, Firm Size variables[50]
2002	46	AR, AZ, DC, MA, MS	No	Yes	No	No
2003	47	AZ, DC, MA, MS	No	Yes	No	No
2004-2008	49	DC, MA	No	Yes	No	No
2009	49	DC, MA	No	Yes	Yes	No
2010	50	MA	Yes	Yes	Yes	No
2011-2016	51	(none)	Yes	Yes	Yes	Yes
2017-2022	50	AK	Yes	Yes	Yes	Yes

job type, and (in the case of the RAC/WAC) segment of the workforce. OD files are also separated by whether the residence location of each job is in the state ("main" files) or outside of the state ("aux" files). This last differentiation allows users to aggregate resident workers for a state without having to touch the much larger "main" files while preventing the duplication of job counts within the dataset.

Released LODES datasets are tagged only with census block codes. Any aggregation to hierarchical geographic areas must be performed by the user with either the LODES block crosswalk or with some other relationship file constructed from the TIGER/Line database: `https://www.census.gov/geographies/reference-files/time-series/geo/relationship-files.html`.

Between the release of LODES Version 5 and LODES Version 6, the base geography in the LODES processing system changed from 2000 tabulation blocks to 2010 tabulation blocks. For data years 2002-2009, the public-use data were not re-synthesized in the new geography base, rather these job counts were transcoded into 2010 geography using the process described in U.S. Census Bureau [2015]. In cases where users prefer to perform their own transcoding of these older years of data, Version 5 remains posted on the LEHD website; however, it will not be updated with newer data than 2009 or with expanded coverage. Newer years of data and other structural changes to LODES will be implemented in Version 7 and higher as needed.

6.5 OnTheMap

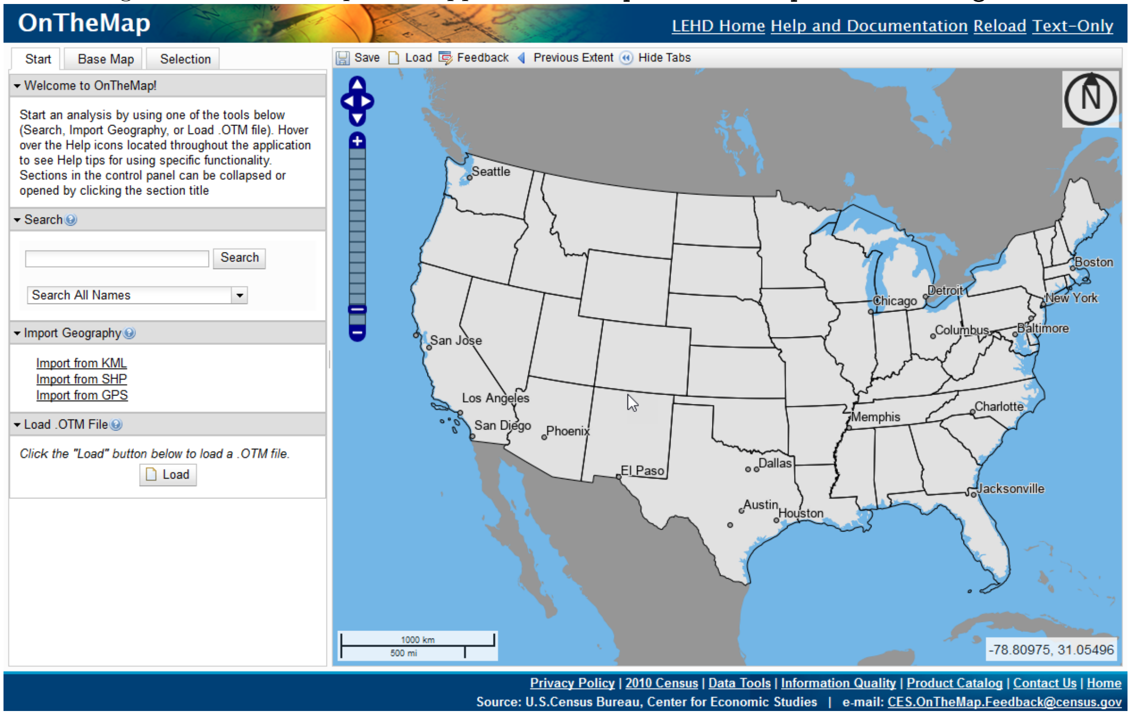
OnTheMap is a free web-mapping and analysis tool for the LODES datasets. OnTheMap provides users with a simple interface to make queries into the (Version 6) LODES data and display summarized results. A screenshot from the OnTheMap Web Application is shown in Figure 10. Additional images of specific queries and output reports are provided in Appendix B.

Files that are fed into the OnTheMap database are processed versions of the public-use data produced by the LODES processing system. In order to make queries within OnTheMap as fast as possible for users, the public data are mixed with a hierarchical geographic crosswalk (relationship file), pre-tabulated, and broken into separate files. These file structures are not exposed directly in OnTheMap but they do directly support the 6 kinds of analyses that are offered to users in the application:

- Area Profile (uses RAC or WAC)
- Area Comparison (uses RAC or WAC)
- Distance/Direction (uses OD)
- Destination (uses OD)
- Inflow/Outflow (uses OD)
- Paired Area (uses OD)

Additionally, OnTheMap only presents 2 of the 3 ownership types (Private and All).

Figure 10: OnTheMap Web Application: https://onthemap.ces.census.gov/



6.6 Block Crosswalk

In order to be able to aggregate the block-based OnTheMap data to higher geographies it is necessary to construct a "block crosswalk." The block crosswalk relates 2010 census blocks to other higher level geography supported by the OnTheMap Application. Thus, through the use of a nationally complete block crosswalk, a specific block can be referenced to determine into which metropolitan area, congressional district, place, etc. it falls. The block crosswalk has one entry for each 2010 census block and one column for each higher geographical layer that is supported.[51]

The block crosswalk is based upon TIGER relationship files as well as some external datasets. These external datasets allow the addition of Workforce Investment Board (WIB) areas, which are not supported in Census Bureau's standard geography base.

The construction of a block crosswalk for the purpose of aggregating data presents a number of methodological issues, primarily centered around the conflict between the 2010 census block universe and the "current" geography in which many other supported layers are defined. A description of the relationship between 2010 geography definitions and current definitions can typically be found in the Census Bureau's TIGER technical documentation.

[51] The current version of LODES (8.4) is based on 2020 Census Block definitions, with a similar methodology.

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A Quality Statistics

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This appendix provides additional detail on edit and imputation statistics, geocoding quality statistics, quality statistics for residential distributions by workplace locations, and transition tables for small workplace and residential cells. Unless otherwise specified, statistics in the section report on LODES data for 2018.

Total Variability of Workplace Job Counts

In McKinney et al. [2020], the authors measure total variability for the core measure used in LODES (beginning of quarter employment) for several tables, or characteristic crossings. While these characteristic crossings align with the public-use QWI statistics, not all of these tables are available in the public-use LODES. Additionally, LODES reports certain characteristics in collapsed categorizations compared with QWI's publication schema (specifically worker age and industry grouping).

Of the characteristic crossings reported in Table 2 of McKinney et al. [2020], only Age×Sex and NAICS Sector×County can be created using the public-use LODES. Table 13 summarizes the public-use LODES data for 2018 in the same form as McKinney et al. [2020] for these two characteristic crossings, providing both the number of cells and the median count (no independent estimate of the total variability is made here). Note that in the first case (Age×Sex), LODES only provides three age groups, as compared to QWI's eight age groups. Additionally, we did not independently estimate the number of sampling zeroes in the NAICS Sector×County crossing. The median counts of jobs in Table 13 are generally consistent with those in McKinney et al. [2020], with reasonable deviations (e.g. the difference in number of age categories).

Table 13: Cell Sizes for Comparison to McKinney et al. [2020]

Table and Employment Count Range	Number of Cells	Median Count
Age×Sex		
1000+	300	314,774.5
NAICS Sector×County		
1-2	1,056	2.0
3-9	2,662	6.0
10-99	15,759	42.0
100-999	24,693	299.0
1000+	15,332	2963.0

Edit and Imputation Statistics

Table 14 shows workplace imputation rates by state and for the entire country. The imputation steps are defined in Section 4.2.2 and in Figure 2. The workplace imputation table shows that most (97.30%) of jobs in the United States did not require the imputation of a workplace location, and state-level imputation rates generally fall between 1% and 4%. When a workplace does need to be imputed, most are handled with Impute 1, which is drawn from workers who have the same residence location, the same industry, and are employed in the same state. Rates for Impute 2 are almost always less than 0.5% and Impute 3 usually less than 0.06%. The QWI imputation of workplaces is used not at all in some states and rarely in the rest of the states. For example, in Alabama, 96.99% of workplaces required no imputation; 2.81% received workplace characteristics from Impute 1; 0.19% received workplace characteristics from Impute 2; 0.02% received workplace characteristics from Impute 3; and no observations received the modal characteristics from the QWI Production System.

Table 14: Workplace Imputations Rates, by State

Emp. State	None [%]	Impute 1 [%]	Impute 2 [%]	Impute 3 [%]	QWI Impute [%]
US	97.29	2.56	0.14	0.01	<0.01
AL	96.99	2.81	0.19	0.02	0
AZ	97.09	2.73	0.16	0.02	0

AR	95.13	4.49	0.34	0.03	0
CA	98.13	1.80	0.07	<0.01	0
CO	98.64	1.29	0.07	<0.01	0
CT	97.82	2.07	0.10	<0.01	0
DE	93.26	6.21	0.50	0.03	0
DC	95.52	4.22	0.25	<0.01	0
FL	96.27	3.59	0.13	<0.01	0
GA	97.37	2.50	0.12	0.01	0
HI	99.39	0.41	0.19	<0.01	0
ID	97.41	2.37	0.17	0.05	0
IL	98.02	1.91	0.06	<0.01	0
IN	97.15	2.72	0.12	<0.01	0
IA	98.75	1.16	0.07	0.01	0
KS	98.05	1.86	0.08	<0.01	0
KY	97.76	2.08	0.15	<0.01	0
LA	97.57	2.25	0.18	<0.01	0
ME	97.48	2.26	0.21	0.05	0
MD	97.60	2.30	0.09	0.01	0
MA	98.43	1.49	0.07	0.01	0
MI	97.08	2.83	0.08	<0.01	0
MN	98.23	1.69	0.06	0.01	0
MS	95.98	3.69	0.30	0.03	0
MO	96.74	3.11	0.14	0.02	0
MT	97.73	1.93	0.27	0.06	0
NE	97.11	2.72	0.15	0.02	0
NV	97.72	2.11	0.15	0.03	0
NH	96.83	2.94	0.18	0.05	0
NJ	97.51	2.37	0.12	0.01	0
NM	97.64	1.97	0.35	0.04	0
NY	97.12	2.76	0.11	<0.01	0
NC	97.92	1.98	0.10	<0.01	0
ND	99.35	0.51	0.11	0.02	0
OH	97.81	2.09	0.09	<0.01	0
OK	98.31	1.51	0.17	<0.01	0
OR	97.71	2.14	0.15	<0.01	0
PA	97.17	2.69	0.13	<0.01	0
RI	97.73	2.09	0.14	0.03	0
SC	95.98	3.78	0.21	0.03	0
SD	99.07	0.82	0.09	0.02	0
TN	96.94	2.92	0.13	0.01	0
TX	95.37	4.30	0.32	0.01	<0.01
UT	98.43	1.47	0.08	0.01	0
VT	96.29	3.24	0.45	0.01	0
VA	95.14	4.61	0.22	0.03	0
WA	99.34	0.63	0.03	<0.01	0
WV	94.63	4.43	0.86	0.08	0
WI	98.01	1.91	0.07	<0.01	0
WY	97.20	2.17	0.59	0.04	0

Table 15 shows residential imputation rates by state and for the entire country. The imputation steps are defined in Section 4.2.2 and in Figure 3. The residence imputation tables shows that most (95.81%) of workers' residences in the United States did not require imputation, and state-level imputation rates generally range from 2% to 10%. The relative usage of imputations #1 and #2 varies significantly by state. Imputation #3 is used much less often, usually less than 0.2%. For example, in Alabama, 95.55% of jobs required no residence imputation; 1.59% received a residence from Impute 1; 2.75% received a residence from Impute 2; and the remaining 0.12% received a residence from Impute 3.

Table 15: Residential Imputations Rates, by State

Res. State	None [%]	Impute 1 [%]	Impute 2 [%]	Impute 3 [%]
US	95.83	0.69	3.37	0.11
AL	95.55	1.59	2.75	0.12
AK	92.60	0.26	7.10	0.03
AZ	96.00	0.41	3.48	0.11
AR	95.14	0.30	4.33	0.23
CA	95.71	0.45	3.77	0.07
CO	95.40	0.58	3.95	0.06
CT	96.43	0.32	3.17	0.07
DE	97.35	0.34	2.13	0.18
DC	97.40	0.04	2.44	0.13
FL	96.87	0.50	2.51	0.11
GA	96.27	0.89	2.76	0.08
HI	72.15	0.45	27.25	0.16
ID	93.86	1.04	4.98	0.12
IL	96.34	0.29	3.31	0.05
IN	96.86	0.22	2.85	0.08
IA	97.32	0.82	1.82	0.04
KS	97.02	0.36	2.56	0.06
KY	94.93	1.77	3.19	0.11
LA	95.06	1.52	3.28	0.14
ME	94.41	0.56	4.90	0.12
MD	97.16	0.38	2.40	0.07
MA	96.13	0.24	3.56	0.06
MI	98.19	0.33	1.43	0.05
MN	97.63	0.39	1.94	0.03
MS	94.68	1.35	3.78	0.19
MO	96.87	0.61	2.43	0.09
MT	91.20	0.86	7.75	0.19
NE	96.56	1.07	2.28	0.09
NV	95.21	0.80	3.89	0.10
NH	95.00	0.80	4.09	0.10
NJ	94.84	0.66	4.40	0.10
NM	90.87	0.73	8.22	0.18
NY	95.71	0.07	4.12	0.10
NC	95.35	0.94	3.63	0.08
ND	94.67	1.06	4.23	0.05
OH	97.85	0.39	1.70	0.06
OK	92.54	2.67	4.65	0.13
OR	95.03	0.62	4.25	0.11
PA	97.52	0.44	1.96	0.08
RI	96.18	0.15	3.60	0.07
SC	96.58	0.94	2.35	0.13
SD	93.51	1.71	4.70	0.07
TN	97.11	0.57	2.24	0.08
TX	94.02	1.72	3.95	0.31
UT	95.01	1.08	3.85	0.06
VT	90.21	0.87	8.55	0.38
VA	96.46	0.55	2.83	0.16
WA	94.58	0.94	4.44	0.04
WV	90.41	3.57	5.56	0.46
WI	97.41	0.42	2.12	0.04
WY	89.45	0.62	9.53	0.40

Geocoding Quality Statistics

As mentioned above, geocoding quality statistics are provided in Table 16. This table reports workplace geocoding quality in several categories by state. The shares are job weighted and the categories are as follows:

Rooftop, Block Certain The workplace address was geocoded to the "rooftop" level using all the address information and the census block is certain. This is the ideal for geocoding addresses within the Geocoded Address List (GAL).

ZIP+4, Block Certain The workplace address was geocoded based on the ZIP+4[52] for the address. Although a "rooftop" designation could not be achieved, the census block of the address is still considered certain.

Blockgroup Based on address information, the Census blockgroup could be ascertained, but not the Census block within the blockgroup.

County Based on the address information, the county of the workplace is all that could be determined.

State The address provided for the workplace could not be geocoded, but because of how the data are collected, we know the state of the workplace.

In Table 16, we can see that the great majority of jobs are at addresses for which the census block is certain and most of those have received a "rooftop" level geocode. Because of the nature of geocoding addresses and how census geography are formed, very few addresses receive a blockgroup-quality geocode. Tract-level is typically the second-largest constributor, but still a small share overall. County and state level geocodes are usually no more than a few percent of the state total. For example, in Alabama, 91.64% of workplaces were geocoded to Rooftop with the census block being certain; an additional 1.68% with a certain block within a USPS ZIP+4 area; less than 0.01% of workplaces only received a census blockgroup assignment; 3.67% receiving only a census tract assignment; 1.21% receiving only a county assignment; and 1.80% receiving no geocoding detail other than state.

There is quite a bit of variance among states. A couple states - Alaska and West Virginia - have noticeably low rooftop geocoding relative to other states. In these cases, Tract level geocoding is higher as well as ZIP+4 (WV only). While city-style addresses are generally easier to geocode with more precision, some cities have address unique addressing features that that can adversely affect geocoding quality. Examples include hyphenated addresses in Queens, NY and the quadrant system in Washington, DC.

Table 16: Workplace Geocoding Quality, by State

	Geocode Quality Status, [% of State Employment]					
	Rooftop, Block Certain	ZIP+4, Block Certain	Blockgroup	Tract	County	State
AL	91.64	1.68	<0.01	3.67	1.21	1.80
AZ	93.93	0.62	0	2.53	2.28	0.63
AR	89.46	1.17	<0.01	4.49	2.94	1.93
CA	95.84	1.04	0.03	1.22	0.96	0.91
CO	96.06	0.87	<0.01	1.70	1.21	0.15
CT	94.56	0.85	0	2.42	1.20	0.97
DE	88.14	1.05	0	4.08	6.04	0.70
DC	91.84	0.48	0.02	3.18	4.48	0
FL	93.95	0.82	<0.01	1.51	2.65	1.08
GA	93.38	1.51	<0.01	2.49	1.27	1.36
HI	94.57	1.08	0.04	3.70	0.60	<0.01
ID	94.04	1.13	0	2.24	1.53	1.06
IL	94.57	1.18	<0.01	2.26	0.55	1.43
IN	94.29	0.89	0	1.97	1.44	1.41
IA	94.74	1.70	0	2.32	1.25	<0.01
KS	94.17	1.43	0	2.46	1.83	0.12
KY	93.31	2.41	0	2.04	1.77	0.47
LA	91.47	1.89	0	4.21	2.26	0.17
ME	94.33	0.33	0	2.82	1.50	1.02
MD	92.92	0.52	<0.01	4.17	0.60	1.80
MA	96.86	0.77	0	0.80	0.67	0.90

[52]See `https://en.wikipedia.org/wiki/ZIP_Code#ZIP.2B4` for more information on the ZIP+4 definition.

MI	93.60	1.02	0	2.46	1.56	1.37
MN	95.47	1.45	0	1.30	0.31	1.46
MS	87.86	2.19	<0.01	5.94	1.74	2.28
MO	92.70	1.71	0	2.32	1.46	1.80
MT	91.89	2.03	0	3.81	1.19	1.07
NE	94.02	1.63	<0.01	1.46	1.78	1.11
NV	95.94	0.38	<0.01	1.40	2.10	0.18
NH	94.04	0.63	0	2.17	0.81	2.35
NJ	89.17	1.75	<0.01	6.59	0.95	1.55
NM	89.49	1.63	<0.01	6.52	1.20	1.15
NY	92.06	1.79	<0.01	3.28	1.92	0.96
NC	92.91	1.16	0	3.84	1.47	0.62
ND	93.95	1.47	0	3.94	0.44	0.21
OH	95.85	0.53	0	1.42	1.46	0.73
OK	92.36	2.22	0	3.73	0.91	0.79
OR	96.17	0.46	0	1.08	1.62	0.67
PA	94.17	1.08	<0.01	1.92	1.45	1.37
RI	94.80	1.06	<0.01	1.87	0.67	1.59
SC	91.14	1.50	0	3.34	2.03	1.99
SD	94.47	1.96	0	2.64	0.69	0.24
TN	94.10	1.40	<0.01	1.44	2.14	0.92
TX	91.03	1.49	0	2.85	4.01	0.62
UT	91.51	3.37	0	3.55	1.57	0
VT	89.39	1.91	0	5.00	3.15	0.56
VA	91.23	1.15	<0.01	2.76	3.60	1.27
WA	96.34	0.75	<0.01	2.25	0.66	<0.01
WV	82.92	4.22	<0.01	7.49	4.77	0.60
WI	95.01	1.36	0	1.63	0.97	1.03
WY	91.09	1.56	0	4.56	2.35	0.45

Additional Quality Measures

Small Cell Transition Tables

We use small cell transition tables to communicate how cells with small counts can be transformed as part of the applied confidentiality protections. The transition tables reported below provide number of cells (job weighted) making the transitions from *confidential count* to *protected count* as a share of the blocks with that confidential count. For both employment and residential blocks, we consider counts binned into the following categories: [*0 jobs*, *1 job*, *2 jobs*, *3 jobs*, *4 jobs*, *5 or more jobs*]. In cases where we have "fractional" jobs due to multiple imputation, the job count is rounded and then binned.

Additional Divergence Measures for Commuting Patterns

In addition to Mean Jensen-Shannon Divergence (JSD) mentioned in Section 5 (Equation 1), we discuss two other commonly used divergence measures: Mean Kullback-Leibler Divergence (KLD) and Root Mean Integrated Square Error (RMISE). Consistent with earlier notation, we calculate P_{ij}, which is the confidential share of jobs with residence grid cell (*distance* $\times$ *direction*), i, as a fraction of all jobs with employment location, j. Correspondingly, Q_{ij} is the protected approximation of P_{ij}. Using these variables, we can define the KLD and the RMISE as follows:

Mean Kullback-Leibler Divergence

$$\bar{D}_{KL}(P||Q) = \frac{1}{N}\sum_{j=1}^{J} n_j \sum_{i=1}^{G} P_{ij} log_2 \frac{P_{ij}}{Q_{ij}} \tag{2}$$

Here, n_j is the count of confidential jobs in workplace, j, and N is the total count of jobs in all workplaces, J. G is the total number of *distance* $\times$ *direction* grid cells, which is 25 (see Table 8 for details).

It should be noted that the Jensen-Shannon Divergence is a symmetric version of the Kullback-Leibler Divergence. In this quality profile, we use the Jensen-Shannon Divergence because it addresses some problematic properties of the KLD for this dataset. First. the KLD is undefined when $Q_{ij} = 0$ and $P_{ij} \neq 0$, which happens frequently when the data are sparse. Additionally, the KLD is unbounded, whereas the JSD is bounded $[0, 1]$ when we use log_2. Finally, the KLD is not symmetric - meaning that $\bar{D}_{KL}(P, Q) \neq \bar{D}_{KL}(Q, P)$ - whereas the JSD from P to Q is the same as the JSD from Q to P.

While the KLD has been used for research into new protection methods, we prefer to publish the JSD instead of the KLD because of these differences.

Root Mean Integrated Square Error

For the purposes of this quality profile, we define the Root Mean Integrated Square Error as follows. The mean square error is aggregated (weighted by the confidential count in each workplace, j) over all workplaces, J.

$$RMISE(P, Q) = \sqrt{\frac{1}{N}\sum_{j=1}^{J} n_j \sum_{i=1}^{29} (P_{ij} - Q_{ij})^2} \tag{3}$$

General Quality Statistics

Small Cell Transition Tables

The following tables show transition matricies between underlying block counts and the published block counts for both workplaces and residences. For example, 92.58% of blocks that actually had one job in them were published with one job, whereas 7.36% of those cells were published with two jobs.

And for residence: Of blocks that had zero workers with a residence in the actual data, 3.79% were published with one worker appearing there. Compare that to 32.00%, which is the share of blocks that actually had one worker's residence in them but were published with zero residences.

Table 17: Employment Block Transition Table [%]

	Protected Count of Jobs in Block					
	0	1	2	3	4	5+
Actual Count of Jobs in Block						
0	100.00	<0.01	<0.01	<0.01	<0.01	<0.01
1	0.06	92.58	7.36	<0.01	<0.01	<0.01
2	<0.01	66.29	31.91	1.60	0.20	<0.01
3	<0.01	2.49	47.98	40.09	8.37	1.07
4	0	0.05	4.54	38.87	39.47	17.08
5+	<0.01	<0.01	0.02	0.50	3.20	96.28

Table 18: Residence Block Transition Table [%]

	Protected Count of Jobs in Block					
	0	1	2	3	4	5+
Actual Count of Jobs in Block						
0	90.26	3.79	2.28	1.31	0.75	1.61
1	33.61	19.15	14.76	10.34	6.87	15.27
2	32.00	18.38	14.66	10.60	7.29	17.07
3	22.70	16.04	14.65	12.02	9.23	25.36
4	14.95	12.42	13.20	12.31	10.53	36.59
5+	2.42	1.69	2.31	2.79	3.13	87.67

Residence Distribution Divergence Measures

Table 19 and 20 display Jensen-Shannon Divergence (JSD) and Root Mean Integrated Square Error (RMISE) for residence distributions in each workplace location. The statistics are aggregated to the national level for reporting here. Additionally, the statistics are reported separately for workplace of different sizes, which shows clearly that small workplaces have higher relative error.

In Table 19 we can see that the national JSD is less than 0.01.

Table 19: Jensen-Shannon Divergence, by State

	Mean Jensen-Shannon Divergence				
	Workplace Tract Size				
	All	$[0, 100)$	$[100, 1000)$	$[1000, 10000)$	$[10000, max)$
US	<0.01	0.09	0.02	<0.01	<0.01

Table 20 gives a similar array of information for the RMISE. Nationally, the RMISE is 0.03 for all tracts and generally close to this value for medium, large, and very large tracts for most states.

Table 20: Root Mean Integrated Squared Error, by State

	Root Mean Integrated Squared Error				
	Workplace Tract Size				
	All	$[0, 100)$	$[100, 1000)$	$[1000, 10000)$	$[10000, max)$
US	0.03	0.20	0.07	0.03	0.01

B Figures

This appendix provides additional figures for reference purposes.

OnTheMap Sample Screenshots

The following figures show screenshots from several different analyses in the OnTheMap application. The screenshots were taken on April 8, 2019 from the public website, `https://onthemap.ces.census.gov/`, which was showing Version 6.6 of the application. The LODES data version was 20170818, and in all cases 2015 data were used.

Figure 11 shows a Work Area Profile Analysis for the city of San Mateo, CA. The query identifies all "Primary" jobs that have their employment location in one of the census blocks associated with San Mateo. Then the distribution of this employment, by block, is shown using larger/darker dots to represent more employment. Charts are tables of the distribution of jobs by business and worker characteristics are also provided, and the employment distribution of these subpopulations can also be mapped.

Figure 12 shows a Distance/Direction Analysis for the city of San Mateo, CA. The query identifies the same set of jobs as in the Work Area Profile Analysis above (Figure 11), but in this case we map the residential distribution of those jobs. The distribution in this case (an option for users) is presented as a job density "heat map". Also shown is the count of jobs by a cross-tabulation of distance and direction measures (8 directions, 4 distances). (Distance and direction are calculated from internal points of the respective residential and workplace census blocks.) Each of the distance groups can be mapped independently.

Figure 13 shows an Inflow/Outflow Analysis for the city of San Mateo, CA. In this case, the query identifies several populations: Jobs with workplaces in San Mateo but with residences outside the city; jobs with residences in San Mateo but workplaces outside the city; and jobs with both workplaces and residences in San Mateo. The totals from these queries and their linear combinations are presented in graphical and tabular form.

Figure 11: OnTheMap: Work Area Profile Analysis for San Mateo, CA

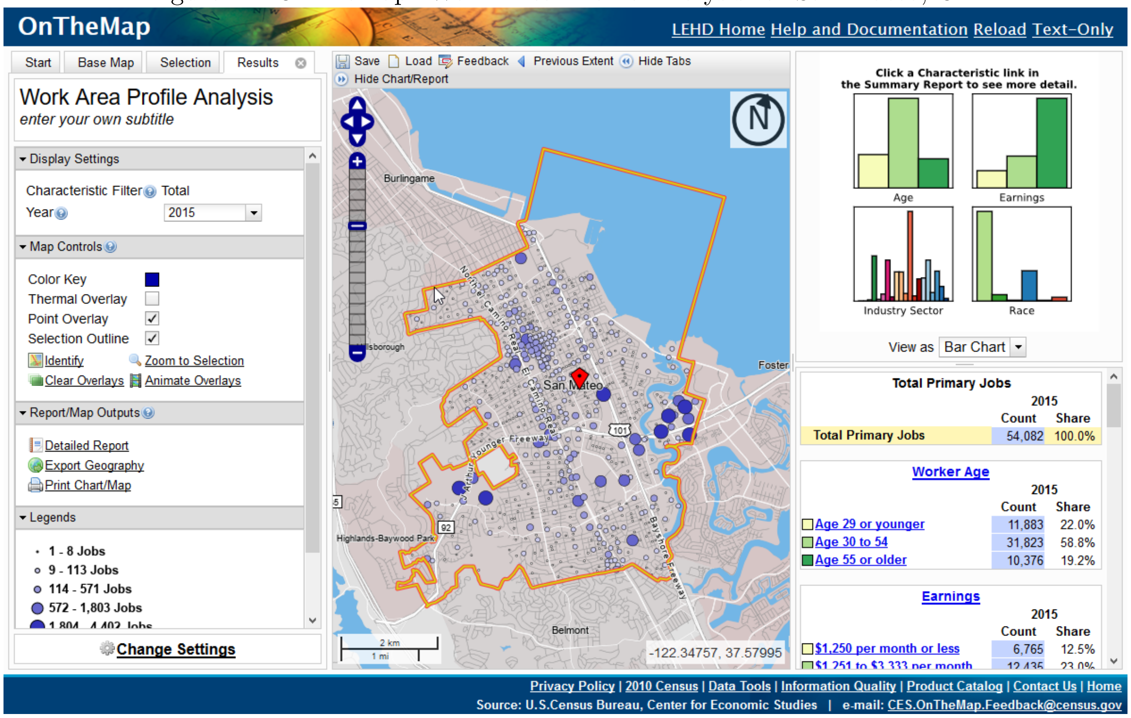


Figure 12: OnTheMap: Distance/Direction Analysis for San Mateo, CA

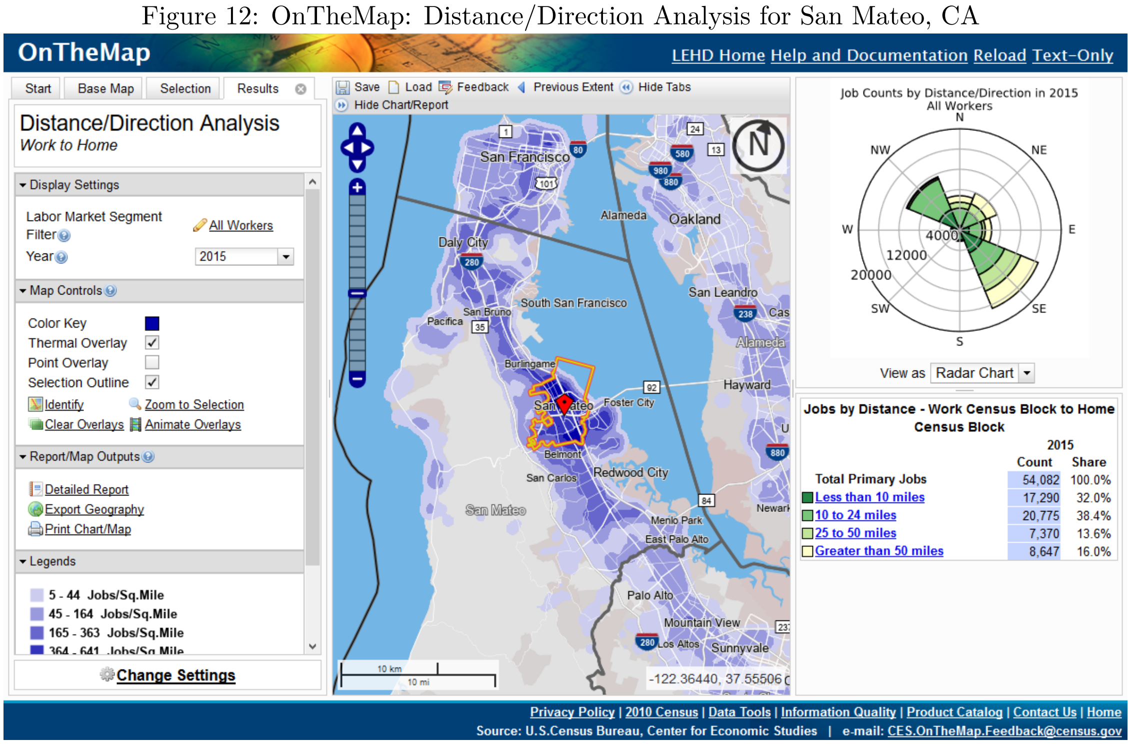
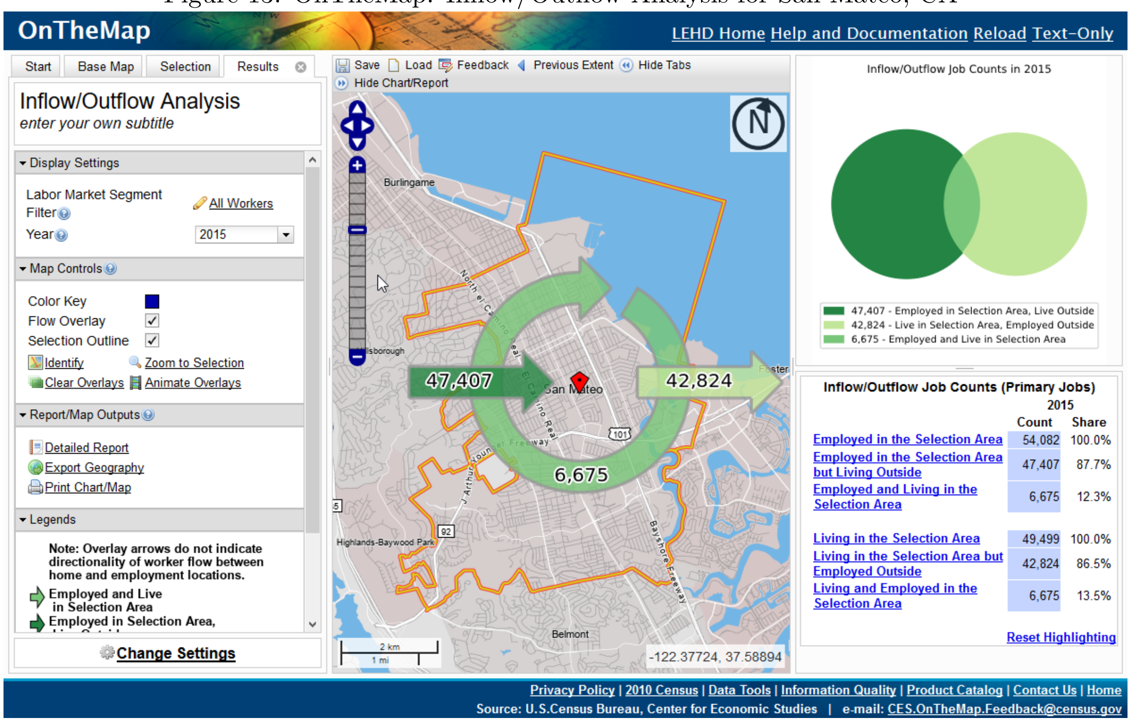


Figure 13: OnTheMap: Inflow/Outflow Analysis for San Mateo, CA



OnTheMap Sample Reports

This section shows example reports from three different analyses: A Work Area Profile analysis; a Destination analysis, and an Inflow/Outflow analysis.

Work Area Profile Report

Total Primary Jobs

	2015 Count	2015 Share
Total Primary Jobs	54,082	100.0%

Jobs by Worker Age

	2015 Count	2015 Share
Age 29 or younger	11,883	22.0%
Age 30 to 54	31,823	58.8%
Age 55 or older	10,376	19.2%

Jobs by Earnings

	2015 Count	2015 Share
$1,250 per month or less	6,765	12.5%
$1,251 to $3,333 per month	12,435	23.0%
More than $3,333 per month	34,882	64.5%

Jobs by NAICS Industry Sector

	2015 Count	2015 Share
Agriculture, Forestry, Fishing and Hunting	4	0.0%
Mining, Quarrying, and Oil and Gas Extraction	0	0.0%
Utilities	174	0.3%
Construction	6,048	11.2%
Manufacturing	246	0.5%
Wholesale Trade	948	1.8%
Retail Trade	5,488	10.1%

Page: 1
Source: U.S. Census Bureau, OnTheMap Application, https://onthemap.ces.census.gov

Jobs by NAICS Industry Sector

	2015 Count	2015 Share
Transportation and Warehousing	570	1.1%
Information	3,910	7.2%
Finance and Insurance	3,897	7.2%
Real Estate and Rental and Leasing	1,028	1.9%
Professional, Scientific, and Technical Services	11,998	22.2%
Management of Companies and Enterprises	706	1.3%
Administration & Support, Waste Management and Remediation	2,979	5.5%
Educational Services	3,097	5.7%
Health Care and Social Assistance	5,488	10.1%
Arts, Entertainment, and Recreation	1,152	2.1%
Accommodation and Food Services	4,016	7.4%
Other Services (excluding Public Administration)	1,964	3.6%
Public Administration	369	0.7%

Jobs by Worker Race

	2015 Count	2015 Share
White Alone	36,682	67.8%
Black or African American Alone	2,658	4.9%

Page: 2
Source: U.S. Census Bureau, OnTheMap Application, https://onthemap.ces.census.gov

Jobs by Worker Race

	2015 Count	2015 Share
American Indian or Alaska Native Alone	385	0.7%
Asian Alone	12,410	22.9%
Native Hawaiian or Other Pacific Islander Alone	353	0.7%
Two or More Race Groups	1,594	2.9%

Jobs by Worker Ethnicity

	2015 Count	2015 Share
Not Hispanic or Latino	44,278	81.9%
Hispanic or Latino	9,804	18.1%

Jobs by Worker Educational Attainment

	2015 Count	2015 Share
Less than high school	4,682	8.7%
High school or equivalent, no college	7,066	13.1%
Some college or Associate degree	11,452	21.2%
Bachelor's degree or advanced degree	18,999	35.1%
Educational attainment not available (workers aged 29 or younger)	11,883	22.0%

Jobs by Worker Sex

	2015 Count	2015 Share
Male	28,929	53.5%

Page: 3

Source: U.S. Census Bureau, OnTheMap Application, https://onthemap.ces.census.gov

Jobs by Worker Sex

	2015 Count	2015 Share
Female	25,153	46.5%

Page: 4

Source: U.S. Census Bureau, OnTheMap Application, https://onthemap.ces.census.gov

Report Settings

Analysis Type	Area Profile
Selection area as	Work
Year(s)	2015
Job Type	Primary Jobs
Labor Market Segment	All Workers
Selection Area	San Mateo city, CA from Places (Cities, CDPs, etc.)
Selected Census Blocks	1,063
Analysis Generation Date	04/08/2019 16:00 - OnTheMap 6.6
Code Revision	862b6296f5ebf0d900479b7d896f6536db69dfe7
LODES Data Version	20170818

Source: U.S. Census Bureau, OnTheMap Application and LEHD Origin-Destination Employment Statistics (Beginning of Quarter Employment, 2nd Quarter of 2002-2015).
Notes:

1. Race, Ethnicity, Educational Attainment, and Sex statistics are beta release results and are not available before 2009.
2. Educational Attainment is only produced for workers aged 30 and over.
3. Firm Age and Firm Size statistics are beta release results for All Private jobs and are not available before 2011.

Source: U.S. Census Bureau, OnTheMap Application, https://onthemap.ces.census.gov

Home Destination Report - Where Workers Live Who are Employed in the Selection Area - by Places (Cities, CDPs, etc.)

Total Primary Jobs

	2015	
	Count	Share
Total Primary Jobs	54,082	100.0%

Jobs Counts by Places (Cities, CDPs, etc.) Where Workers Live - Primary Jobs

	2015	
	Count	Share
San Mateo city, CA	6,675	12.3%
San Francisco city, CA	5,653	10.5%
San Jose city, CA	3,095	5.7%
Redwood City city, CA	1,828	3.4%
Foster City city, CA	1,456	2.7%
Daly City city, CA	1,357	2.5%
Hayward city, CA	1,249	2.3%
South San Francisco city, CA	1,219	2.3%
San Bruno city, CA	1,120	2.1%
Fremont city, CA	1,103	2.0%
All Other Locations	29,327	54.2%

Page: 1

Source: U.S. Census Bureau, OnTheMap Application, https://onthemap.ces.census.gov

Report Settings

Analysis Type	Destination
Destination Type	Places (Cities, CDPs, etc.)
Selection area as	Work
Year(s)	2015
Job Type	Primary Jobs
Selection Area	San Mateo city, CA from Places (Cities, CDPs, etc.)
Selected Census Blocks	1,063
Analysis Generation Date	04/08/2019 15:59 - OnTheMap 6.6
Code Revision	862b6296f5ebf0d900479b7d896f6536db69dfe7
LODES Data Version	20170818

Source: U.S. Census Bureau, OnTheMap Application and LEHD Origin-Destination Employment Statistics (Beginning of Quarter Employment, 2nd Quarter of 2002-2015).
Notes:

1. Race, Ethnicity, Educational Attainment, and Sex statistics are beta release results and are not available before 2009.
2. Educational Attainment is only produced for workers aged 30 and over.
3. Firm Age and Firm Size statistics are beta release results for All Private jobs and are not available before 2011.

Source: U.S. Census Bureau, OnTheMap Application, https://onthemap.ces.census.gov

Inflow/Outflow Report

Selection Area Labor Market Size (Primary Jobs)

	2015	
	Count	**Share**
Employed in the Selection Area	54,082	100.0%
Living in the Selection Area	49,499	91.5%
Net Job Inflow (+) or Outflow (-)	4,583	-

In-Area Labor Force Efficiency (Primary Jobs)

	2015	
	Count	**Share**
Living in the Selection Area	49,499	100.0%
Living and Employed in the Selection Area	6,675	13.5%
Living in the Selection Area but Employed Outside	42,824	86.5%

In-Area Employment Efficiency (Primary Jobs)

	2015	
	Count	**Share**
Employed in the Selection Area	54,082	100.0%
Employed and Living in the Selection Area	6,675	12.3%
Employed in the Selection Area but Living Outside	47,407	87.7%

Page: 1

Source: U.S. Census Bureau, OnTheMap Application, https://onthemap.ces.census.gov

Outflow Job Characteristics (Primary Jobs)

	2015	
	Count	**Share**
External Jobs Filled by Residents	42,824	100.0%
Workers Aged 29 or younger	7,374	17.2%
Workers Aged 30 to 54	25,817	60.3%
Workers Aged 55 or older	9,633	22.5%
Workers Earning $1,250 per month or less	4,770	11.1%
Workers Earning $1,251 to $3,333 per month	9,058	21.2%
Workers Earning More than $3,333 per month	28,996	67.7%
Workers in the "Goods Producing" Industry Class	5,025	11.7%
Workers in the "Trade, Transportation, and Utilities" Industry Class	7,644	17.8%
Workers in the "All Other Services" Industry Class	30,155	70.4%

Inflow Job Characteristics (Primary Jobs)

	2015	
	Count	**Share**
Internal Jobs Filled by Outside Workers	47,407	100.0%
Workers Aged 29 or younger	10,592	22.3%
Workers Aged 30 to 54	28,169	59.4%
Workers Aged 55 or older	8,646	18.2%
Workers Earning $1,250 per month or less	5,601	11.8%

Page: 2

Source: U.S. Census Bureau, OnTheMap Application, https://onthemap.ces.census.gov

Inflow Job Characteristics (Primary Jobs)

	2015	
	Count	Share
Workers Earning $1,251 to $3,333 per month	10,660	22.5%
Workers Earning More than $3,333 per month	31,146	65.7%
Workers in the "Goods Producing" Industry Class	5,907	12.5%
Workers in the "Trade, Transportation, and Utilities" Industry Class	6,331	13.4%
Workers in the "All Other Services" Industry Class	35,169	74.2%

Interior Flow Job Characteristics (Primary Jobs)

	2015	
	Count	Share
Internal Jobs Filled by Residents	6,675	100.0%
Workers Aged 29 or younger	1,291	19.3%
Workers Aged 30 to 54	3,654	54.7%
Workers Aged 55 or older	1,730	25.9%
Workers Earning $1,250 per month or less	1,164	17.4%
Workers Earning $1,251 to $3,333 per month	1,775	26.6%
Workers Earning More than $3,333 per month	3,736	56.0%
Workers in the "Goods Producing" Industry Class	391	5.9%

Page: 3
Source: U.S. Census Bureau, OnTheMap Application, https://onthemap.ces.census.gov

Interior Flow Job Characteristics (Primary Jobs)

	2015	
	Count	Share
Workers in the "Trade, Transportation, and Utilities" Industry Class	849	12.7%
Workers in the "All Other Services" Industry Class	5,435	81.4%

Page: 4
Source: U.S. Census Bureau, OnTheMap Application, https://onthemap.ces.census.gov

Report Settings

Analysis Type	Inflow/Outflow
Selection area as	N/A
Year(s)	2015
Job Type	Primary Jobs
Selection Area	San Mateo city, CA from Places (Cities, CDPs, etc.)
Selected Census Blocks	1,063
Analysis Generation Date	04/08/2019 15:57 - OnTheMap 6.6
Code Revision	862b6296f5ebf0d900479b7d896f6536db69dfe7
LODES Data Version	20170818

Source: U.S. Census Bureau, OnTheMap Application and LEHD Origin-Destination Employment Statistics (Beginning of Quarter Employment, 2nd Quarter of 2002-2015).
Notes:

1. Race, Ethnicity, Educational Attainment, and Sex statistics are beta release results and are not available before 2009.
2. Educational Attainment is only produced for workers aged 30 and over.
3. Firm Age and Firm Size statistics are beta release results for All Private jobs and are not available before 2011.

Source: U.S. Census Bureau, OnTheMap Application, https://onthemap.ces.census.gov

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